GRAND CANYON NATIONAL PARK
WORLD HERITAGE SITE
REHABILITATE NORTH RIM INFRASTRUCTURE
VOLUME 3 – PIPELINES – NORTH RIM TO SUPAI TUNNEL

VOLUME #	DESCRIPTION
VOLUME 0	GENERAL
VOLUME 1	PIPELINES - NORTH RIM
VOLUME 2	PIPELINES - NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 3	PIPELINES - NORTH RIM TO SUPAI TUNNEL
VOLUME 4	PIPELINES - ROARING SPRINGS CAVE TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 5	ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 6	NORTH RIM WATER TREATMENT PLANT
VOLUME 7	NOT USED
VOLUME 8	NORTH RIM WASTEWATER COLLECTION SYSTEM
VOLUME 9	POWER DISTRIBUTION - NORTH RIM
VOLUME 10	POWER DISTRIBUTION - NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 11	POWER DISTRIBUTION - ROARING SPRINGS PUMPHOUSE (RSPH) TO MANZANITA
VOLUME 12	POWER SYSTEM - COTTONWOOD CAMPGROUND

SHEET	SUB SHEET	SUB AREA	TITLE OF SHEET
1	030G001	GENERAL	SHEET INDEX
2	030C101	GENERAL	SITE PLAN
3	030D101	GENERAL	SUPAI FW HYDRAULIC PROFILE AND VALVE SCHEDULE
4	031X101	FW PIPELINE	SITE DEMOLITION PLAN
5	031C101	FW PIPELINE	GESC PLAN
6	031C102	FW PIPELINE	SHEET MAP
7	031C103	FW PIPELINE	PLAN AND PROFILE STA 301+00 TO STA 301+50
8	031C104	FW PIPELINE	PLAN AND PROFILE STA 301+50 TO STA 306+95
9	031C105	FW PIPELINE	PLAN AND PROFILE STA 306+95 TO STA 312+04
10	031C106	FW PIPELINE	PLAN AND PROFILE STA 312+04 TO STA 320+71
11	031C107	FW PIPELINE	PLAN AND PROFILE STA 320+71 TO STA 329+00
12	031C108	FW PIPELINE	PLAN AND PROFILE STA 329+00 TO STA 322+09
13	031C401	FW PIPELINE	SUPAI REST AREA PRV ENLARGED PLAN
14	031C402	FW PIPELINE	SUPAI REST AREA WATER FILLING STATIONS ENLARGED PLAN
15	031C501	FW PIPELINE	WATER SPIGOT AND BOTTLE FILLING STATION DETAILS
16	031C502	FW PIPELINE	AIR RELIEF VALVE VAULT ASSEMBLY PLAN AND SECTIONS
17	031C503	FW PIPELINE	RESTHOUSE VAULT ASSEMBLY PLAN AND SECTIONS
18	031C901	FW PIPELINE	PHOTO DETAILS
19	031S501	FW PIPELINE	PIPELINE ANCHOR DETAILS

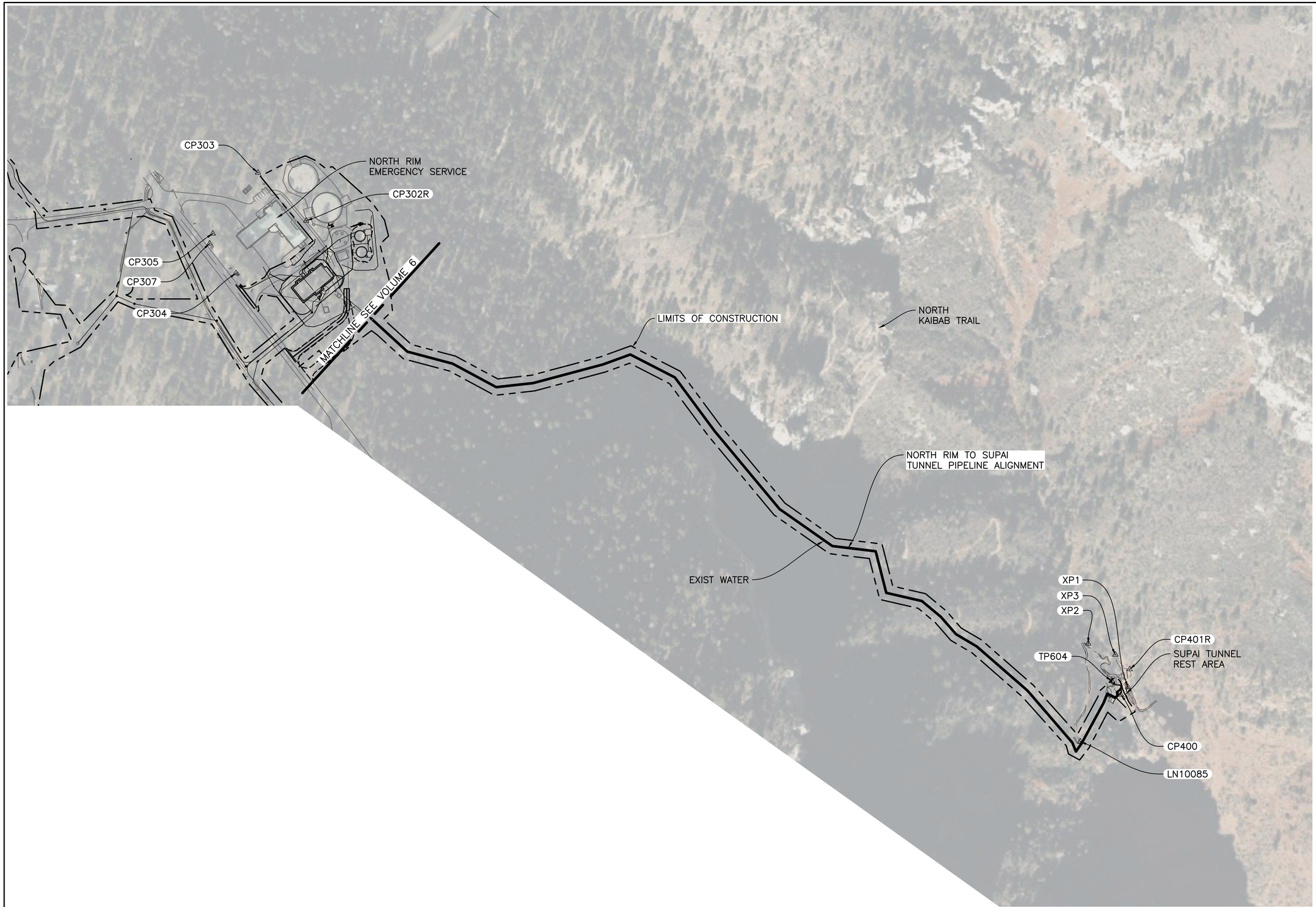
Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY Date Signed 5-1-2026 ARIZONA, U.S.A.

Nicholas Humphrey

CERTIFICATE NO. 41909 RYAN RHOADES Date Signed 5-1-26 REGISTERED PROFESSIONAL ENGINEER ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. HUMPHREY CADD N. LYNCH TECH. REVIEW: M. GOUGH DATE: AUG 2026	SUB SHEET NO. 030 G001	TITLE OF SHEET GENERAL SHEET INDEX NORTH RIM - GRAND CANYON NATIONAL PARK
			DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 1 OF 19

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8/7/2026 1:27:04 PM



GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

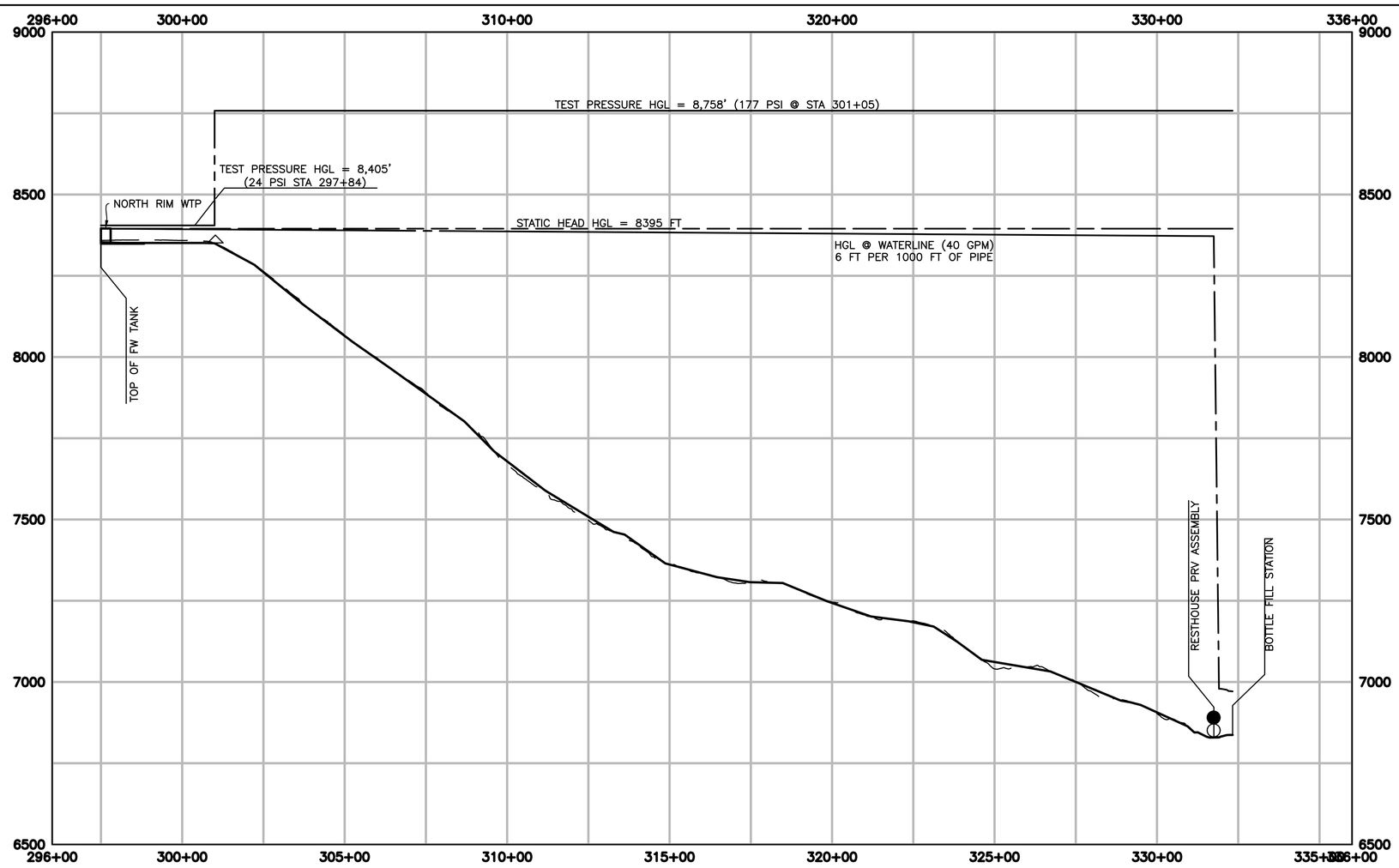
B. SURVEY CONTROL AND CONTROL POINT TABLE LOCATED IN VOLUME 0. SEE SHEETS 002C103 THROUGH 002C105.

150 0 150 300
SCALE OF FEET
1"=150'

Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. HUMPHREY CADD C. WITTENMEIER TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 030 C101	TITLE OF SHEET GENERAL SITE PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 2 OF 19

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HORIZONTAL STATIONING (FEET)

NORTH RIM WATER TREATMENT PLANT TO SUPAI TUNNEL BOTTLE FILL STATION

STATION	LOCATION
297+84	NORTH RIM WTP
331+75	RESTHOUSE PRV ASSEMBLY
332+33	BOTTLE FILL STATION

NORTH RIM WATER TREATMENT PLANT TO SUPAI TUNNEL PIPELINE LOCATIONS

LEGEND:

- COMBINATION AIR RELEASE AND VACUUM VALVE
- ISOLATION VALVE
- PRESSURE REDUCING VALVE
- EXISTING GROUND AT SURFACE
- PIPE CENTERLINE
- WATERLINE HGL
- TEST PRESSURE
- STATIC HEAD

GENERAL NOTES:

A. CONTRACTOR TO SUBMIT TESTING PLAN TO COR FOR REVIEW PRIOR TO TESTING OF THE STATIC LINE

B. FOR FULL PIPELINE LENGTH HYDRAULIC PROFILES SEE VOLUME 0 SHEET 005D103.

200 0 200 400
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. HUMPHREY CADD C. WITTENMEIER TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 030 D101	TITLE OF SHEET GENERAL SUPAI FW HYDRAULIC PROFLE AND VALVE SCHEDULE NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 3 OF 19

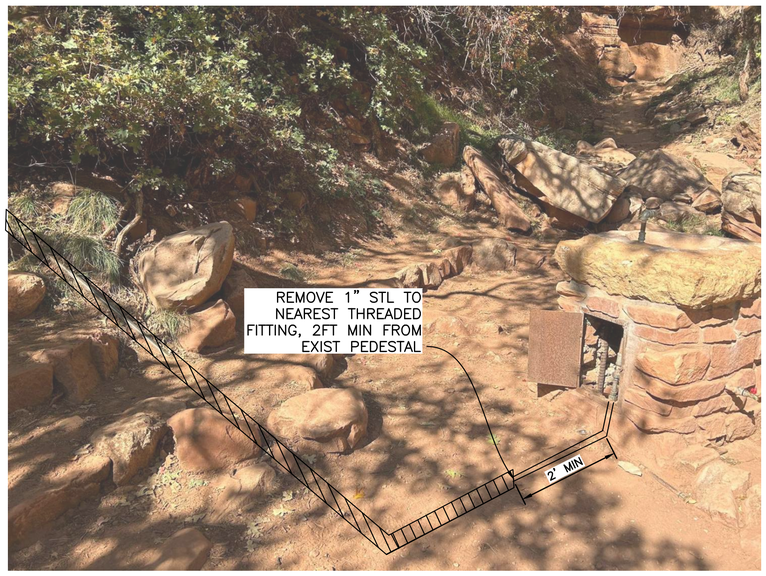
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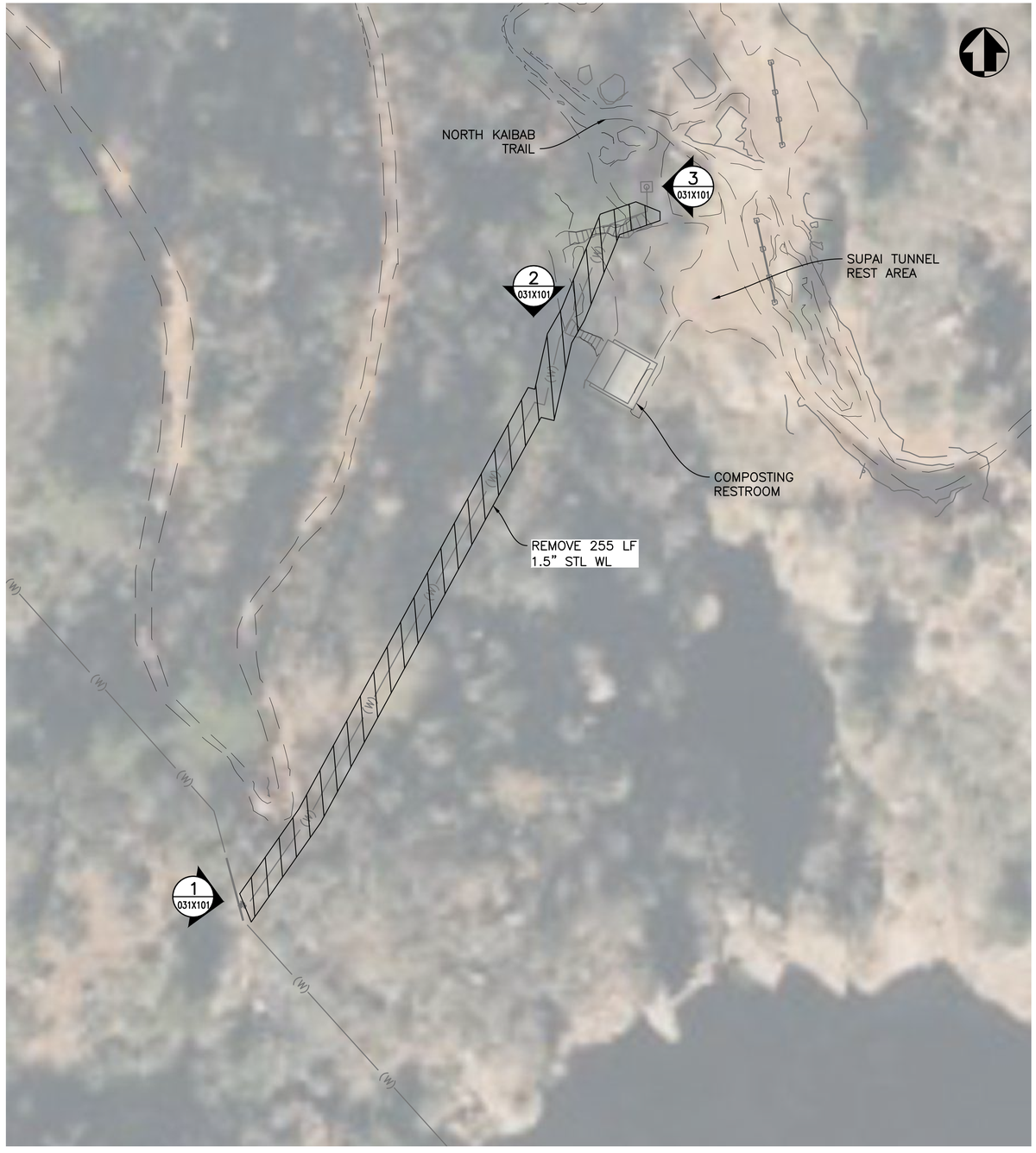
1 031X101 CAP AND ABANDON AT SERVICE TEE
NOT TO SCALE



2 031X101 SERVICE PIPING DEMO AT COMPOSTING RESTROOM
NOT TO SCALE



3 031X101 SUPAI SERVICE PIPING DEMO EXTENTS
NOT TO SCALE

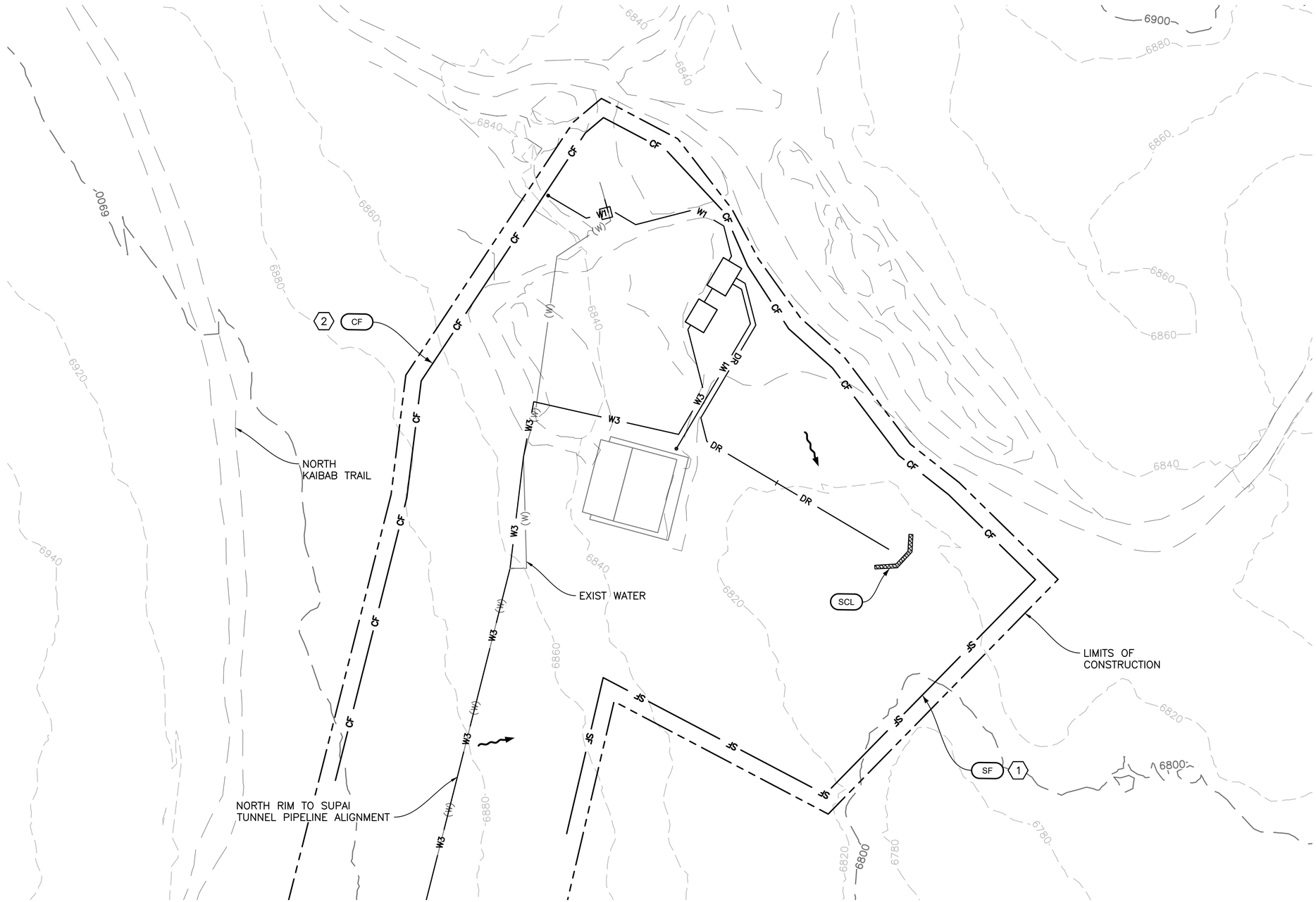


20 0 20 40
SCALE OF FEET
1"=20'

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. HUMPHREY CADD C. WITTENMEIER TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 031 X101	TITLE OF SHEET FW PIPELINE SITE DEMOLITION PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 4 OF 19

4/16/2026 Willyard, Lisa C:\Users\LWILLYARD\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\031-X101.dwg



GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. FOR EROSION CONTROL NOTES AND DETAILS SEE VOLUME 0 SHEETS 002C508 – 002C512

KEYNOTES: (X)

1. SILT FENCE
   1.1. QUANTITY = 133 LF (VOLUME 3 TOTAL)
2. CONSTRUCTION FENCE
   2.1. QUANTITY = 257 LF (VOLUME 3 TOTAL)

LEGEND:

- (SF) — SF — SILT FENCE. SEE 01/002C510
- (CF) — CF — CONSTRUCTION FENCE. SEE 01/002C509
- (SCL) ▨ — SEDIMENT CONTROL LOG. SEE 05/002C509
- FLOW ARROW

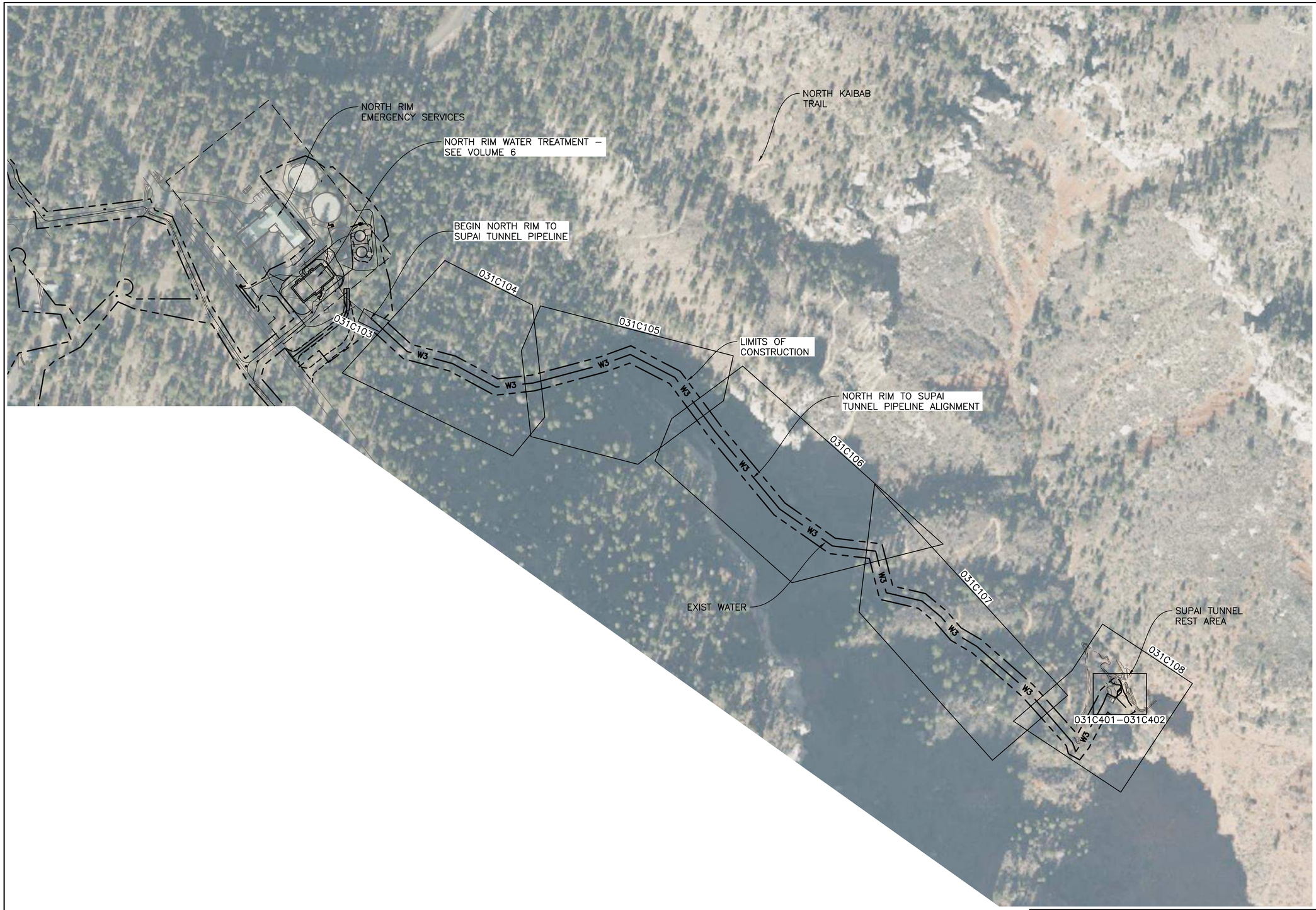
10 0 10 20
SCALE OF FEET
1" = 10'

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. HUMPHREY DRAWN: L. WILLYARD TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 031 C101	TITLE OF SHEET FW PIPELINE GESC PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 5 OF 19

4/16/2026 Willyard, Lisa C:\Users\LWILLYARD\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\031-C101.dwg



GENERAL NOTES

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

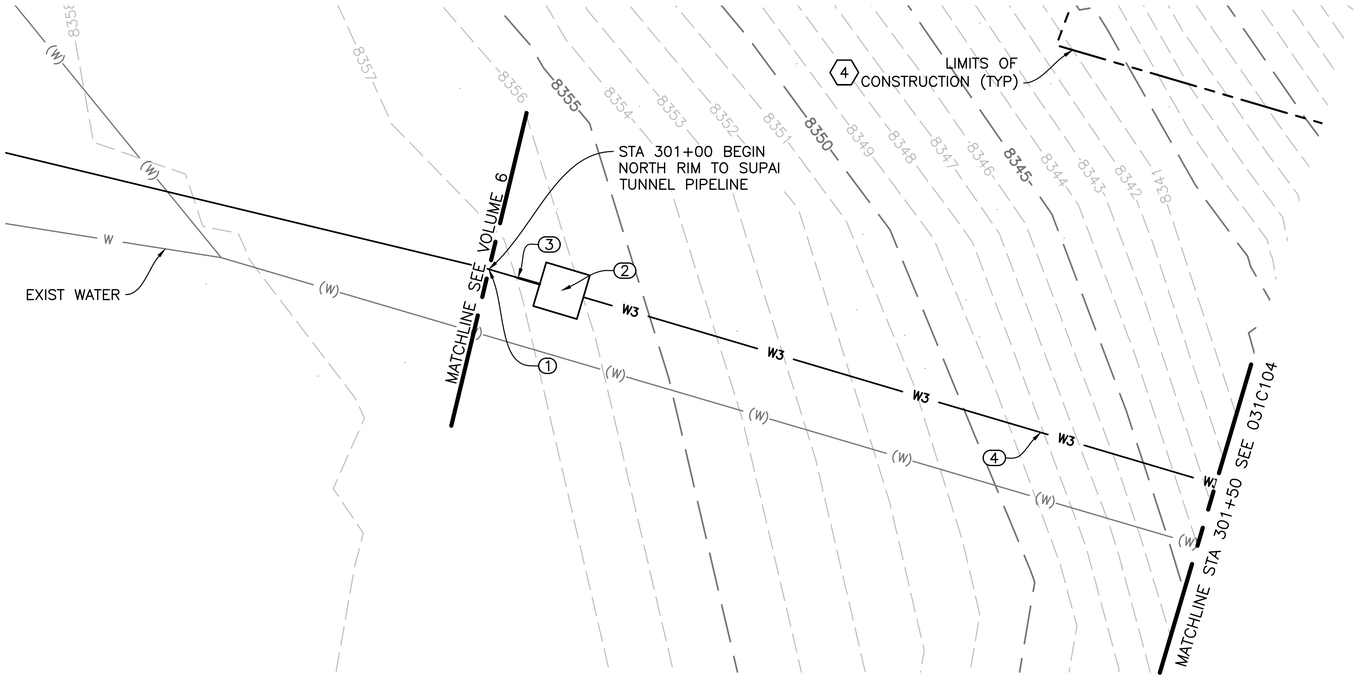
150 0 150 300
SCALE OF FEET
1"=150'

Registered Professional Engineer (Civil)
CERTIFICATE NO.
76139
NICHOLAS
HUMPHREY
5-1-2026
Date Signed
ARIZONA, U.S.A.

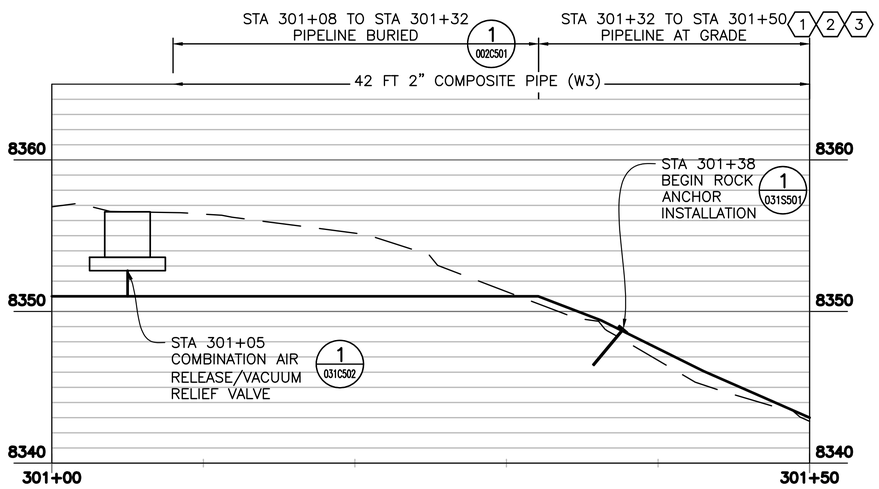
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. HUMPHREY DRAWN: C. WITTENMEIER TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 031 C102	TITLE OF SHEET FW PIPELINE SHEET MAP NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 6 OF 19

4/17/2026 Doll, Abby C:\Users\adoll\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\031-C102.dwg



LOCATION TABLE – NORTH RIM TO SUPAI TUNNEL			
POINT NO.	NORTHING	EASTING	DESCRIPTION
1	13150450.38	1328463.87	CONNECT TO VOL 6 SITE PIPING. SEE SHEET 061C805
2	13150447.01	1328467.57	CENTER OF COMBINATION AIR/VAC VALVE VAULT
3	13150449.03	1328465.35	AIR VENT – CONNECT TO COMBINATION AIR/VAC VALVE VAULT
4	13150424.75	1328491.91	BEGIN ROCK ANCHOR INSTALLATION



GENERAL NOTES

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTOURS AND PROFILE DERIVED FROM REDUCED LIDAR IMAGERY. GROUND DATA IS SHOWN TO PROVIDE A VISUAL REPRESENTATION OF THE CHANGE IN ELEVATION ACROSS THE ALIGNMENT.

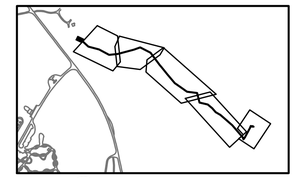
C. EXISTING NORTH RIM TO SUPAI TUNNEL PIPELINE ALIGNMENT IS APPROXIMATE.

D. LENGTHS OF PIPE SHOWN IN PROFILE ARE CALCULATED SLOPE LENGTHS.

KEYNOTES: (X)

1. INSTALL 2" COMPOSITE PIPE AT A 4' OFFSET FROM THE EXISTING 3.5" STEEL WATERLINE.
2. INSTALL PIPE ANCHORS AT A MAXIMUM SPACING OF 275 FT (SLOPE LENGTH). INSTALL PIPE ANCHORS FROM STATION 301+38 TO 332+00 PER DETAIL 01/031S501.
3. CONTRACTOR TO INSTALL DEFLECTION ANCHORS AS REQUIRED TO MAINTAIN PIPE ALIGNMENT PER DETAIL 03/031C501. DO NOT INSTALL COMPOSITE PIPE WITH BEND RADIUS LESS THAN MIN OPERATING BEND RADIUS PER MFR.
4. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTH LEFT AND RIGHT ARE BASED ON WATERLINE ALIGNMENT LOOKING UPSTATION.

CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
301+00	301+30	SEE VOLUME 6 LIMITS OF CONSTRUCTION	
301+30	301+50	25 FT	25 FT



KEYMAP

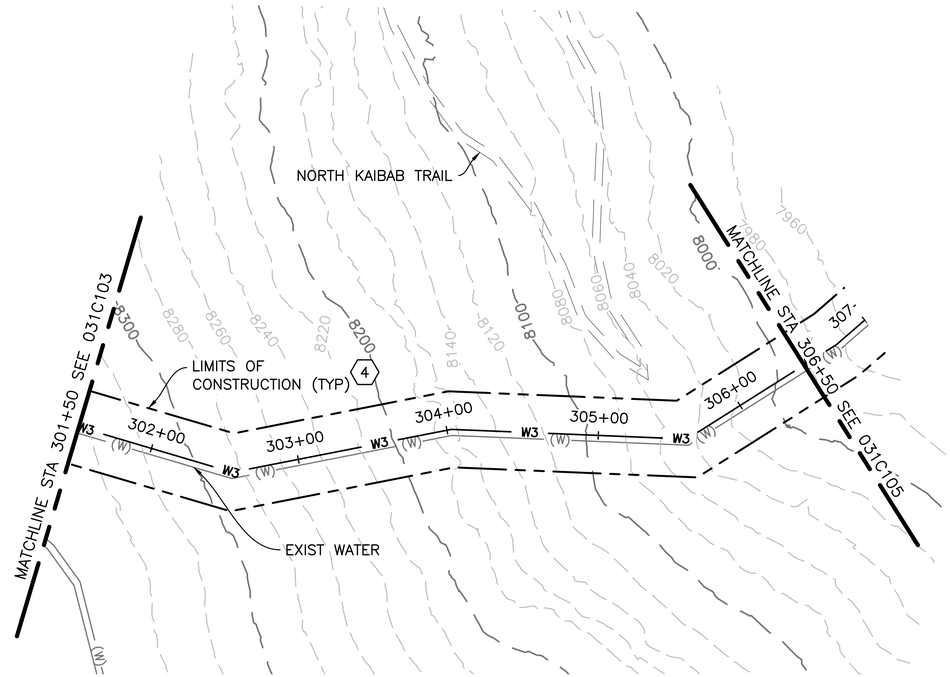
5 0 5 10 HORIZ.
5 0 5 10 VERT.
SCALE OF FEET
H: 1"=5' V: 1"=5'

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

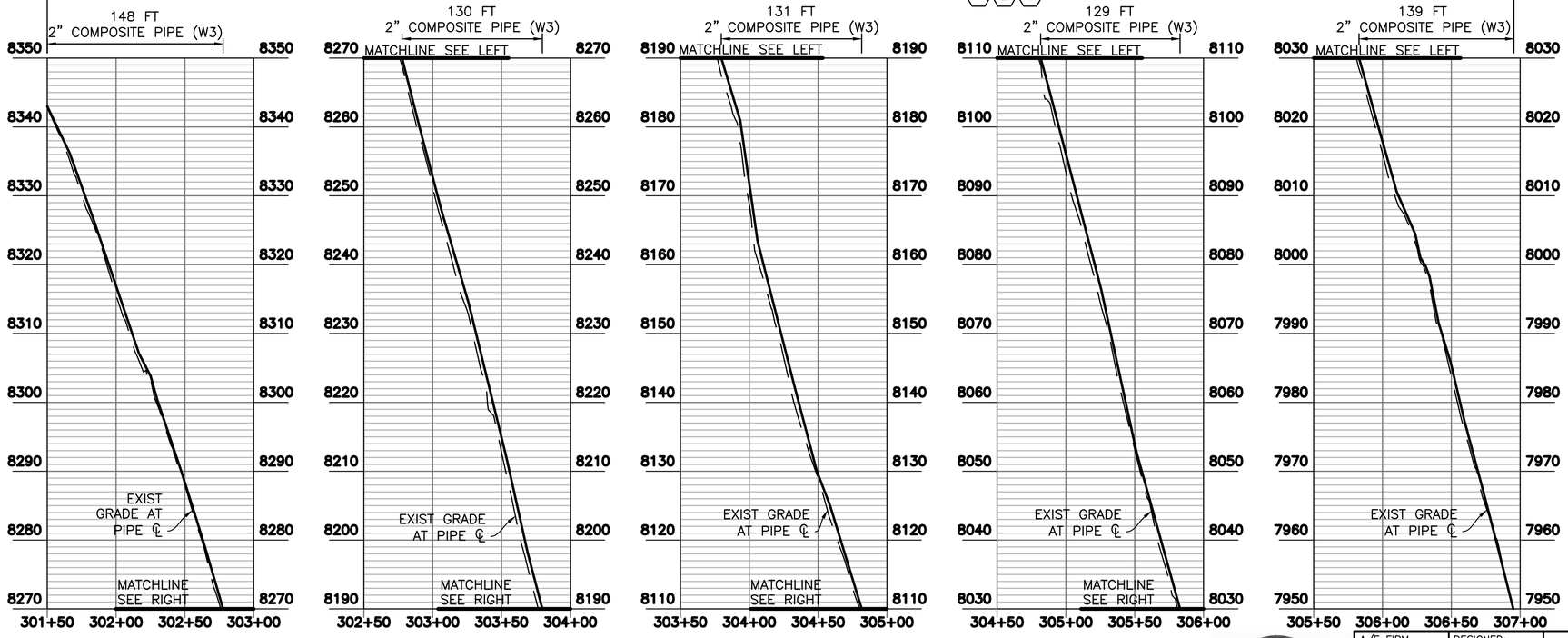
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: M. HUMPHREY DRAWN: C. WITTENMEIER TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 031 C103	TITLE OF SHEET FW PIPELINE PLAN AND PROFILE STA 301+00 TO STA 301+50 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 7 OF 19

4/16/2026 Willyard, Lisa C:\Users\LWILLYARD\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\031-C103.dwg



STA 301+50 TO STA 306+95
PIPELINE AT GRADE 1 2 3



GENERAL NOTES

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTOURS AND PROFILE DERIVED FROM REDUCED LIDAR IMAGERY. GROUND DATA IS SHOWN TO PROVIDE A VISUAL REPRESENTATION OF THE CHANGE IN ELEVATION ACROSS THE ALIGNMENT.

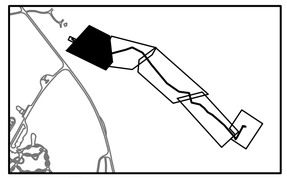
C. EXISTING NORTH RIM TO SUPAI TUNNEL PIPELINE ALIGNMENT IS APPROXIMATE.

D. LENGTHS OF PIPE SHOWN IN PROFILE ARE CALCULATED SLOPE LENGTHS.

KEYNOTES: (X)

1. INSTALL 2" COMPOSITE PIPE AT A 4' OFFSET FROM THE EXISTING 3.5" STEEL WATERLINE.
2. INSTALL PIPE ANCHORS AT A MAXIMUM SPACING OF 275 FT (SLOPE LENGTH). INSTALL PIPE ANCHORS FROM STATION 301+38 TO 332+00 PER DETAIL 01/031S501.
3. CONTRACTOR TO INSTALL DEFLECTION ANCHORS AS REQUIRED TO MAINTAIN PIPE ALIGNMENT PER DETAIL 03/031C501. DO NOT INSTALL COMPOSITE PIPE WITH BEND RADIUS LESS THAN MIN OPERATING BEND RADIUS PER MFR.
4. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTH LEFT AND RIGHT ARE BASED ON WATERLINE ALIGNMENT LOOKING UPSTATION.

CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
301+50	306+50	25 FT	25 FT



KEYMAP

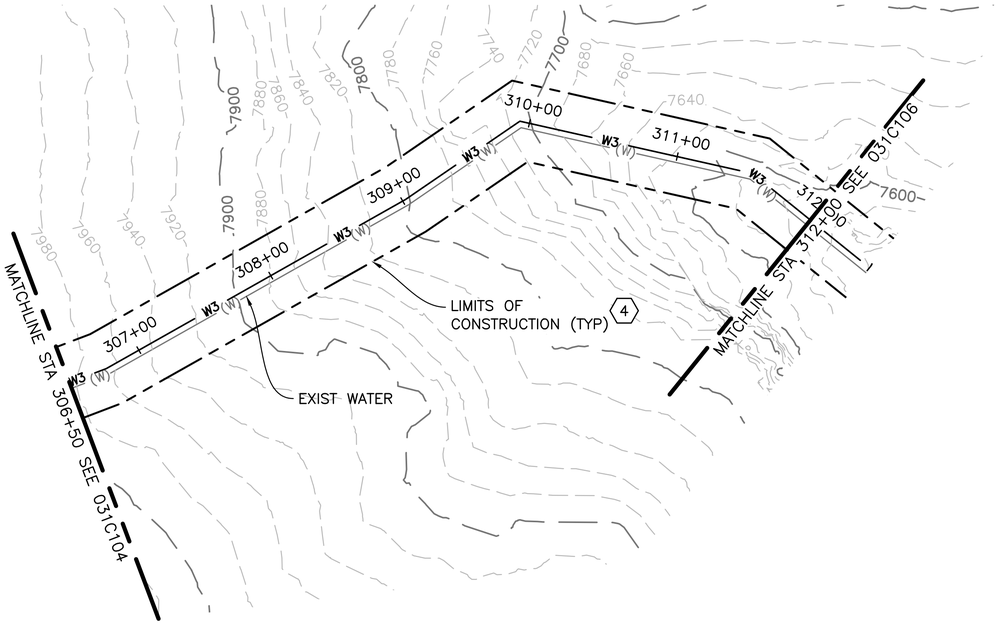
50 0 50 100 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET
H: 1"=50' V: 1"=10'

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

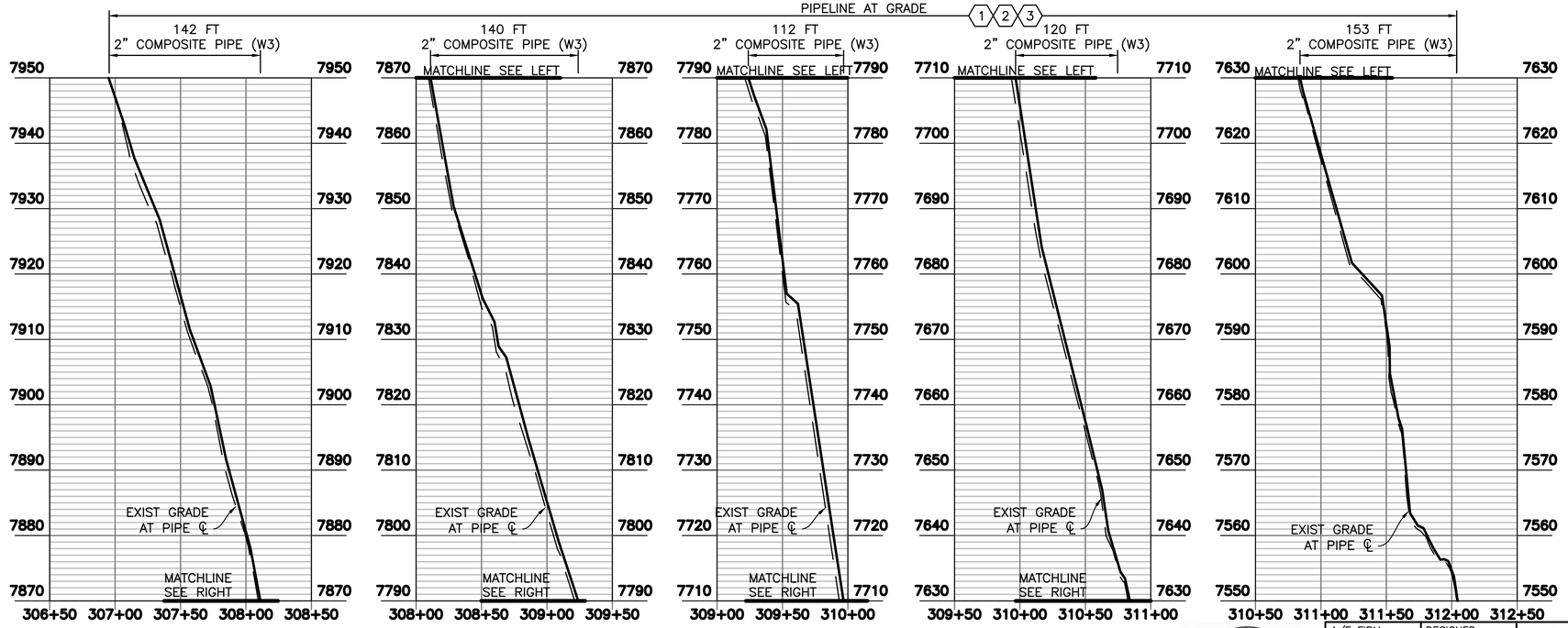
Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	N. HUMPHREY	031 C104	FW PIPELINE PLAN AND PROFILE STA 301+50 TO 306+95	113 182752
	CADD C. WITTENMEIER			PMIS/PKG NO. 318719
	TECH. REVIEW: GOUGH/LUSTER			SHEET 8 OF 19
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	

4/16/2026 Willyard, Lisa C:\Users\LWILLYARD\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\031-C104.dwg



STA 306+95 TO STA 312+04
PIPELINE AT GRADE 1 2 3



GENERAL NOTES

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTOURS AND PROFILE DERIVED FROM REDUCED LIDAR IMAGERY. GROUND DATA IS SHOWN TO PROVIDE A VISUAL REPRESENTATION OF THE CHANGE IN ELEVATION ACROSS THE ALIGNMENT.

C. EXISTING NORTH RIM TO SUPAI TUNNEL PIPELINE ALIGNMENT IS APPROXIMATE.

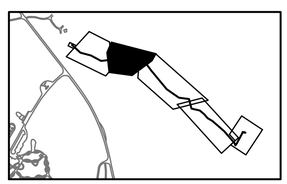
D. LENGTHS OF PIPE SHOWN IN PROFILE ARE CALCULATED SLOPE LENGTHS.

KEYNOTES: (X)

1. INSTALL 2" COMPOSITE PIPE AT A 4' OFFSET FROM THE EXISTING 3.5" STEEL WATERLINE.
2. INSTALL PIPE ANCHORS AT A MAXIMUM SPACING OF 275 FT (SLOPE LENGTH). INSTALL PIPE ANCHORS FROM STATION 301+38 TO 332+00 PER DETAIL 01/031S501.
3. CONTRACTOR TO INSTALL DEFLECTION ANCHORS AS REQUIRED TO MAINTAIN PIPE ALIGNMENT PER DETAIL 03/031C501. DO NOT INSTALL COMPOSITE PIPE WITH BEND RADIUS LESS THAN MIN OPERATING BEND RADIUS PER MFR.
4. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTH LEFT AND RIGHT ARE BASED ON WATERLINE ALIGNMENT LOOKING UPSTATION.

CONSTRUCTION LIMITS TABLE

STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
306+50	312+00	25 FT	25 FT



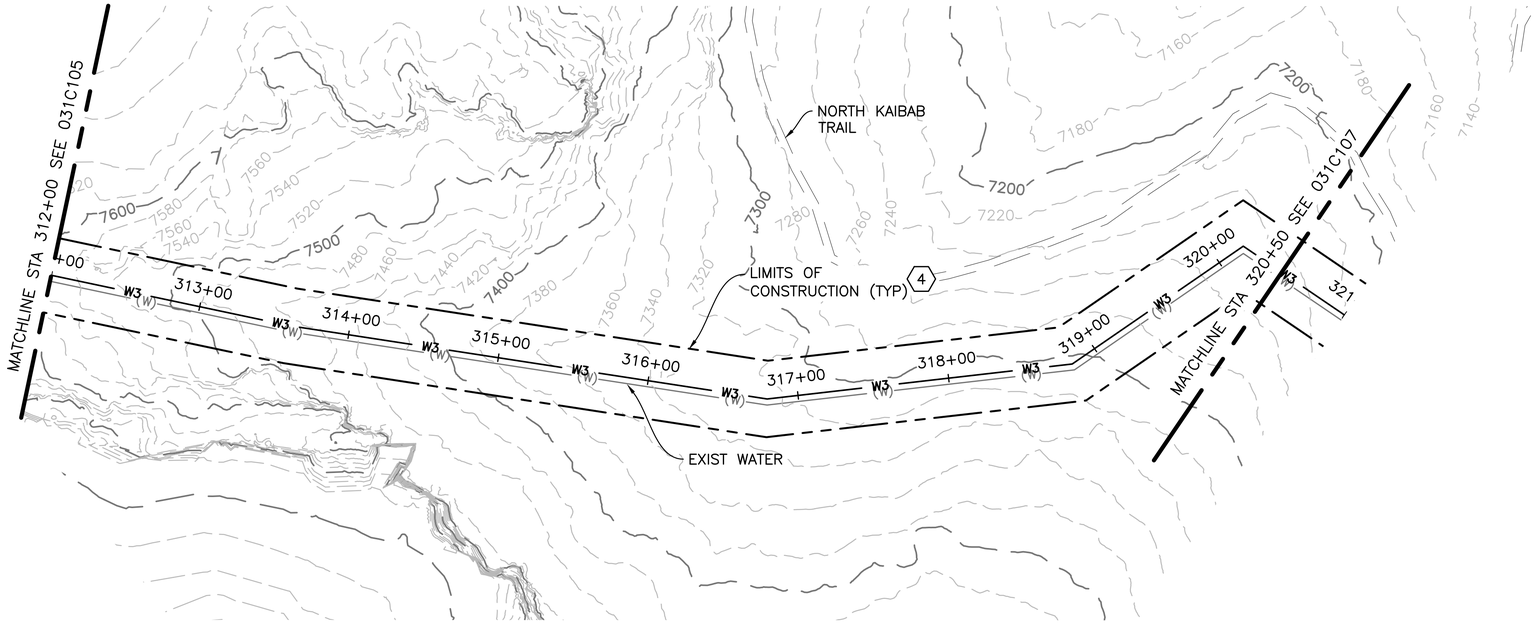
KEYMAP

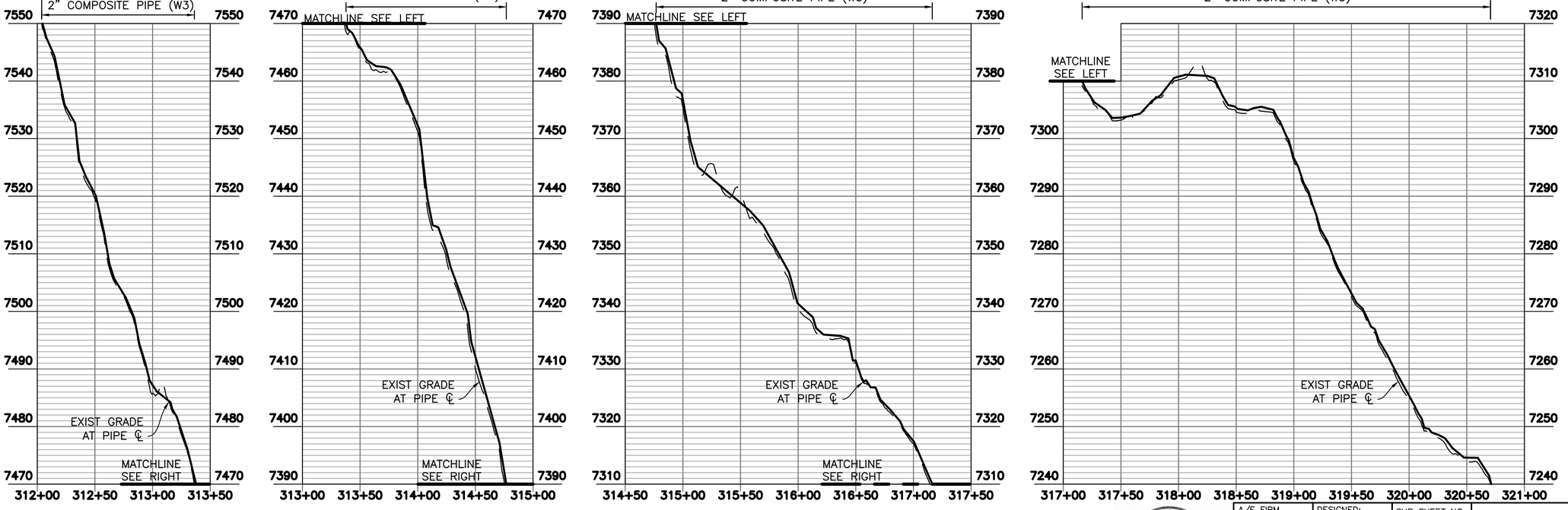
50 0 50 100 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET
H: 1"=50' V: 1"=10'

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 9-1-2026 Not Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM	
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	
DESIGNED:	N. HUMPHREY
CADD	C. WITTENMEIER
TECH. REVIEW:	GOUGH/LUSTER
DATE:	MAY 2026
SUB SHEET NO.	031 C105
TITLE OF SHEET	FW PIPELINE PLAN AND PROFILE STA 306+95 TO 312+04
	NORTH RIM - GRAND CANYON NATIONAL PARK
DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	9 OF 19

4/16/2026 Willyard, Lisa C:\Users\LWILLYARD\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\031-C104.dwg





GENERAL NOTES

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTOURS AND PROFILE DERIVED FROM REDUCED LIDAR IMAGERY. GROUND DATA IS SHOWN TO PROVIDE A VISUAL REPRESENTATION OF THE CHANGE IN ELEVATION ACROSS THE ALIGNMENT.

C. EXISTING NORTH RIM TO SUPAI TUNNEL PIPELINE ALIGNMENT IS APPROXIMATE.

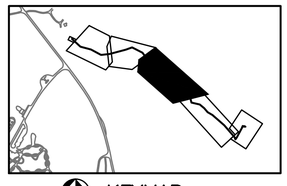
D. LENGTHS OF PIPE SHOWN IN PROFILE ARE CALCULATED SLOPE LENGTHS.

KEYNOTES: ⟨X⟩

1. INSTALL 2" COMPOSITE PIPE AT A 4' OFFSET FROM THE EXISTING 3.5" STEEL WATERLINE.
2. INSTALL PIPE ANCHORS AT A MAXIMUM SPACING OF 275 FT (SLOPE LENGTH). INSTALL PIPE ANCHORS FROM STATION 301+38 TO 332+00 PER DETAIL 01/031S501.
3. CONTRACTOR TO INSTALL DEFLECTION ANCHORS AS REQUIRED TO MAINTAIN PIPE ALIGNMENT PER DETAIL 03/031C501. DO NOT INSTALL COMPOSITE PIPE WITH BEND RADIUS LESS THAN MIN OPERATING BEND RADIUS PER MFR.
4. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTH LEFT AND RIGHT ARE BASED ON WATERLINE ALIGNMENT LOOKING UPSTATION.

CONSTRUCTION LIMITS TABLE

STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
312+00	320+50	25 FT	25 FT



KEYMAP

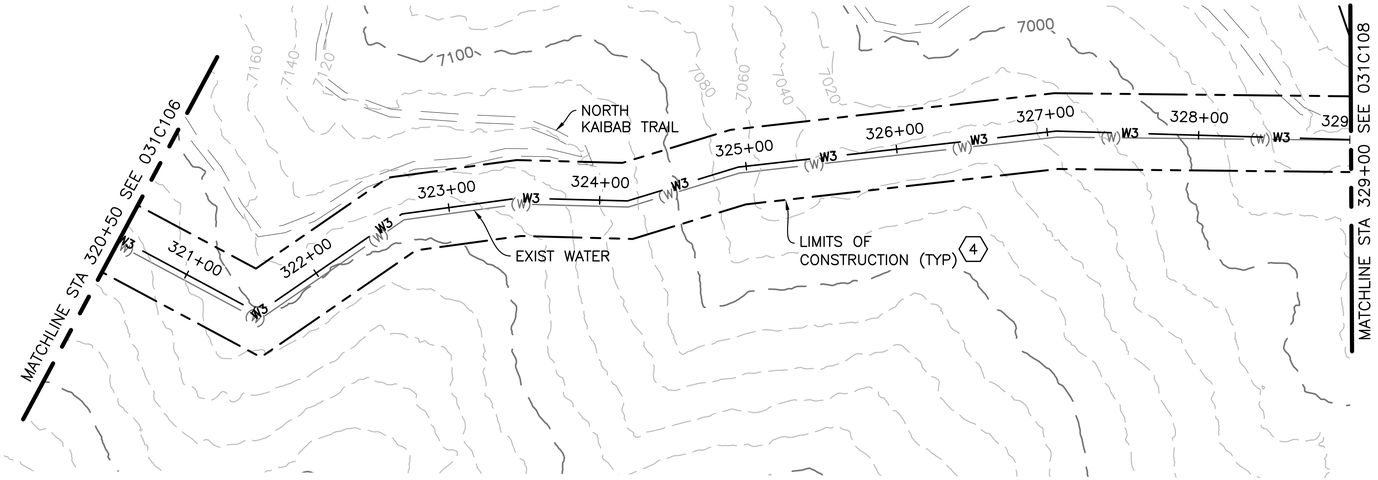
50 0 50 100 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET
H: 1"=50' V: 1"=10'

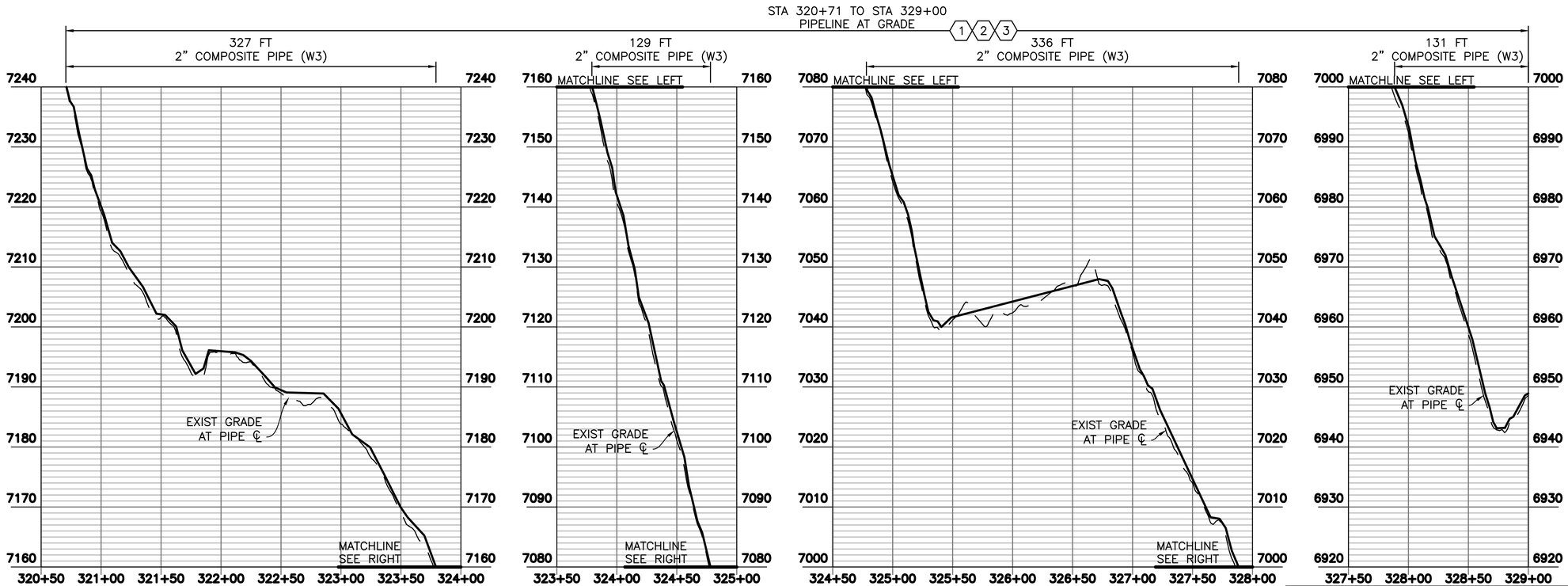
Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	N. HUMPHREY CADD C. WITTENMEIER TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	031 C106	FW PIPELINE PLAN AND PROFILE STA 312+04 TO 320+71 NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 10 OF 19

4/16/2026 Willyard, Lisa C:\Users\LWILLYARD\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\031-C104.dwg





GENERAL NOTES

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTOURS AND PROFILE DERIVED FROM REDUCED LIDAR IMAGERY. GROUND DATA IS SHOWN TO PROVIDE A VISUAL REPRESENTATION OF THE CHANGE IN ELEVATION ACROSS THE ALIGNMENT.

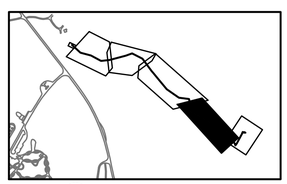
C. EXISTING NORTH RIM TO SUPAI TUNNEL PIPELINE ALIGNMENT IS APPROXIMATE.

D. LENGTHS OF PIPE SHOWN IN PROFILE ARE CALCULATED SLOPE LENGTHS.

KEYNOTES: ⟨X⟩

1. INSTALL 2" COMPOSITE PIPE AT A 4' OFFSET FROM THE EXISTING 3.5" STEEL WATERLINE.
2. INSTALL PIPE ANCHORS AT A MAXIMUM SPACING OF 275 FT (SLOPE LENGTH). INSTALL PIPE ANCHORS FROM STATION 301+38 TO 332+00 PER DETAIL 01/031S501.
3. CONTRACTOR TO INSTALL DEFLECTION ANCHORS AS REQUIRED TO MAINTAIN PIPE ALIGNMENT PER DETAIL 03/031C501. DO NOT INSTALL COMPOSITE PIPE WITH BEND RADIUS LESS THAN MIN OPERATING BEND RADIUS PER MFR.
4. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTH LEFT AND RIGHT ARE BASED ON WATERLINE ALIGNMENT LOOKING UPSTATION.

CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
320+50	329+00	25 FT	25 FT



KEYMAP

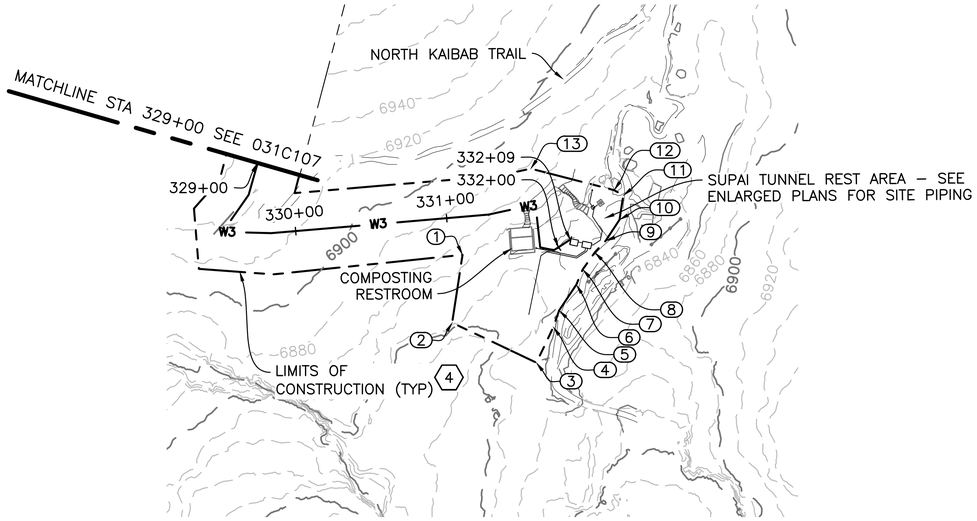
50 0 50 100 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET
H: 1"=50' V: 1"=10'

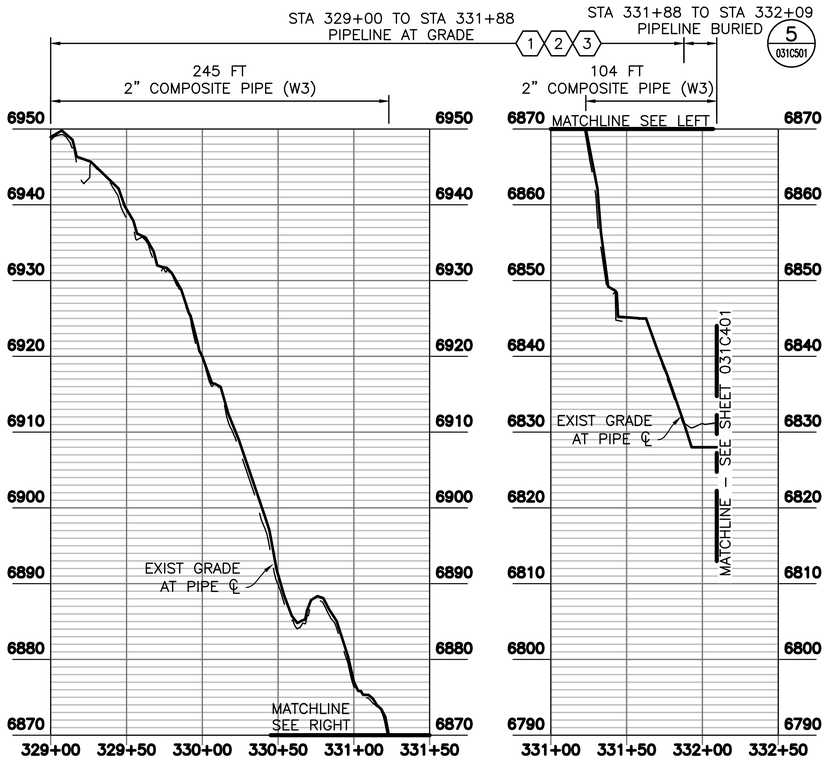
Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. HUMPHREY CADD C. WITTENMEIER TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 031 C107	TITLE OF SHEET FW PIPELINE PLAN AND PROFILE STA 320+71 TO 329+00 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 11 OF 19

4/16/2026 Willyard, Lisa C:\Users\LWILLYARD\DCVACCDocs\HDR\103505545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\031-C104.dwg





GENERAL NOTES

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTOURS AND PROFILE DERIVED FROM REDUCED LIDAR IMAGERY. GROUND DATA IS SHOWN TO PROVIDE A VISUAL REPRESENTATION OF THE CHANGE IN ELEVATION ACROSS THE ALIGNMENT.

C. EXISTING NORTH RIM TO SUPAI TUNNEL PIPELINE ALIGNMENT IS APPROXIMATE.

D. LENGTHS OF PIPE SHOWN IN PROFILE ARE CALCULATED SLOPE LENGTHS.

KEYNOTES: (X)

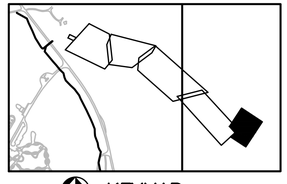
1. INSTALL 2" COMPOSITE PIPE IN SAME ALIGNMENT AS REMOVED 1.5" STEEL PIPE.
2. INSTALL PIPE ANCHORS AT A MAXIMUM SPACING OF 275 FT (SLOPE LENGTH). INSTALL PIPE ANCHORS FROM STATION 301+38 TO 332+00 PER DETAIL 01/031S501.
3. CONTRACTOR TO INSTALL DEFLECTION ANCHORS AS REQUIRED TO MAINTAIN PIPE ALIGNMENT PER DETAIL 03/031C501. DO NOT INSTALL COMPOSITE PIPE WITH BEND RADIUS LESS THAN MIN OPERATING BEND RADIUS PER MFR.
4. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTH LEFT AND RIGHT ARE BASED ON WATERLINE ALIGNMENT LOOKING UPSTATION.

CONSTRUCTION LIMITS TABLE

STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
329+00	331+10	25 FT	25 FT
331+10	332+09	SEE SUPAI REST AREA CONSTRUCTION LIMITS TABLE	

SUPAI REST AREA CONSTRUCTION LIMITS TABLE

POINT NO.	NORTHING	EASTING
1	13149212.99	1330779.44
2	13149182.59	1330813.06
3	13149214.51	1330865.70
4	13149236.71	1330852.24
5	13149245.96	1330845.15
6	13149264.78	1330837.24
7	13149273.94	1330831.13
8	13149286.73	1330826.18
9	13149296.91	1330823.72
10	13149312.19	1330817.17
11	13149323.06	1330805.17
12	13149320.05	1330798.96
13	13149281.91	1330756.81



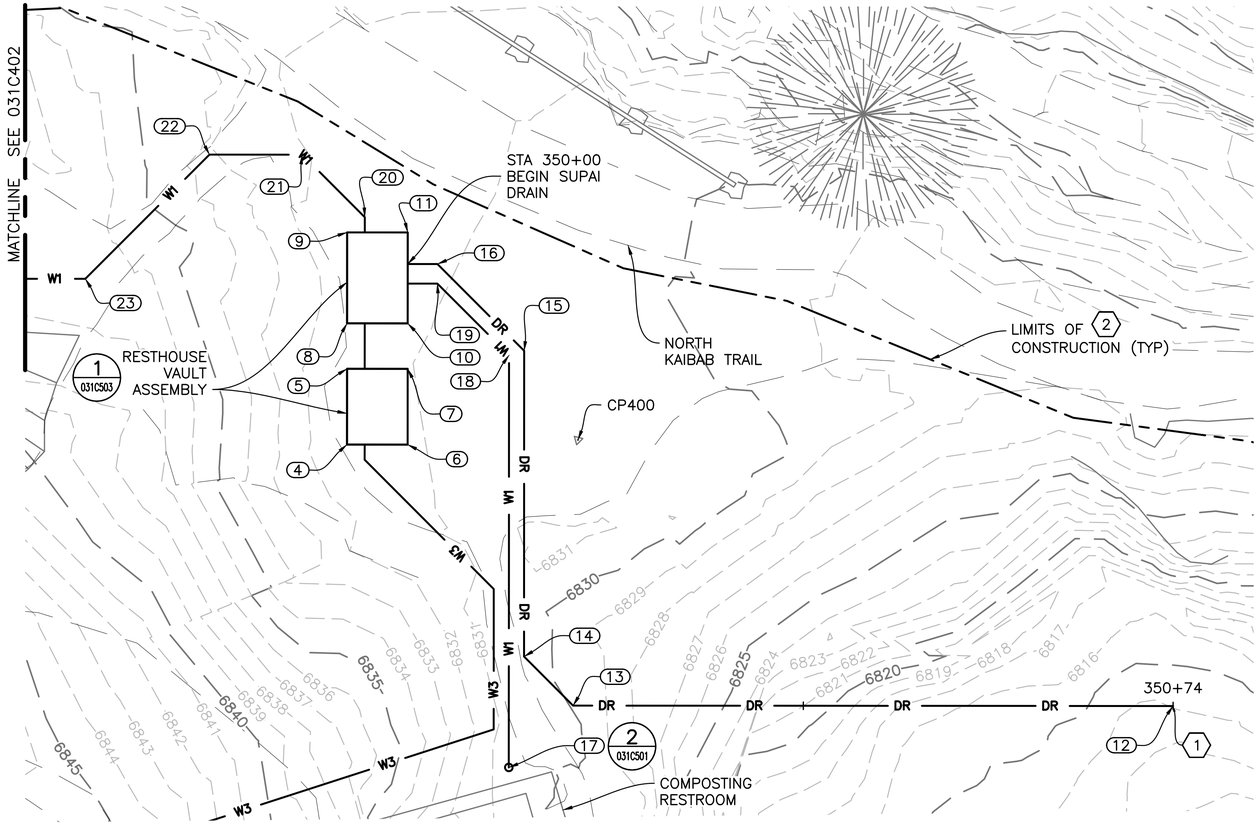
KEYMAP

50 0 50 100 HORIZ.
10 0 10 20 VERT.
SCALE OF FEET
H: 1"=50' V: 1"=10'

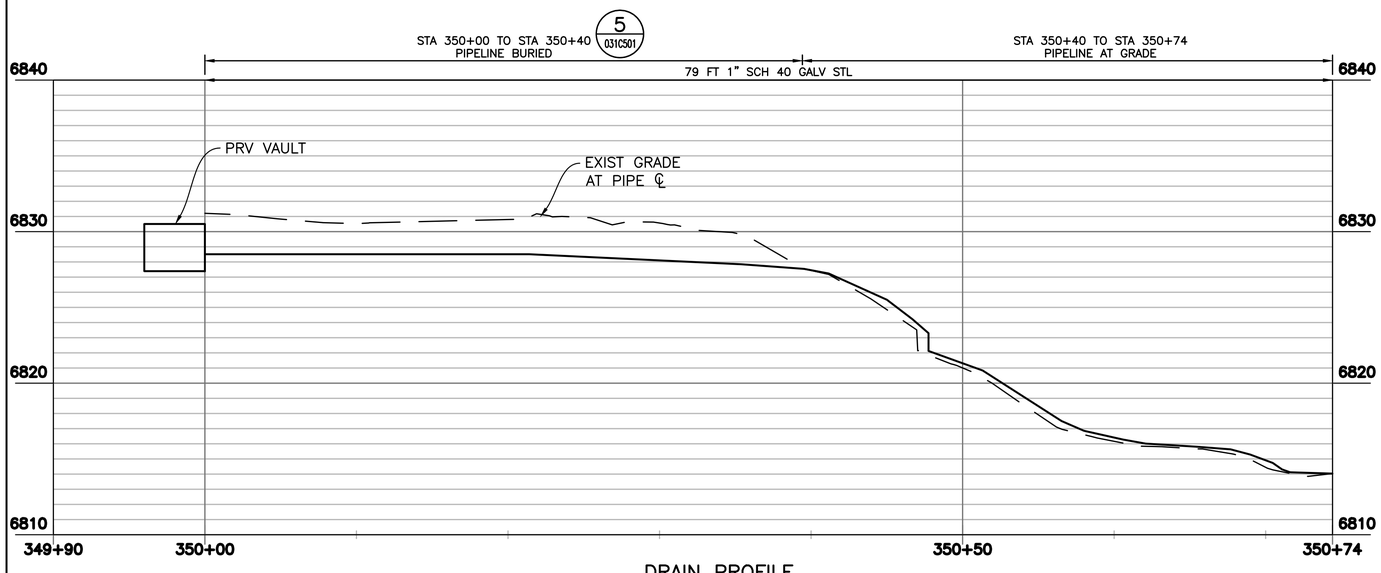
Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. HUMPHREY DRAWN: C. WITTENMEIER TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 031 C108	TITLE OF SHEET FW PIPELINE PLAN AND PROFILE STA 329+00 TO 332+09 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 12 OF 19

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PLAN



DRAIN PROFILE

LOCATION TABLE – SUPAI DISTRIBUTION			
POINT NO.	NORTHING	EASTING	DESCRIPTION
4	13149278.61	1330810.01	VAULT CORNER
5	13149282.14	1330813.55	VAULT CORNER
6	13149275.78	1330812.84	VAULT CORNER
7	13149279.31	1330816.37	VAULT CORNER
8	13149284.26	1330815.67	VAULT CORNER
9	13149288.50	1330819.92	VAULT CORNER
10	13149281.43	1330818.50	VAULT CORNER
11	13149285.67	1330822.74	VAULT CORNER
12	13149227.86	1330836.36	DRAIN TO DAYLIGHT. MAINTAIN MIN 1% SLOPE
13	13149255.88	1330808.38	1" 45° BEND
14	13149260.46	1330808.38	1" 45° BEND
15	13149274.71	1330822.65	1" 45° BEND
16	13149282.81	1330822.65	1" 45° BEND
17	13149255.99	1330802.49	INSTALL WATER FILLING STATION
18	13149275.21	1330821.74	1" 45° BEND
19	13149281.90	1330821.74	1" 45° BEND
20	13149288.38	1330821.45	1" 45° BEND
21	13149294.22	1330821.45	1" 45° BEND
22	13149298.56	1330817.12	1" 45° BEND
23	13149298.56	1330805.53	1" 45° BEND

GENERAL NOTES

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. SURVEY CONTROL AND CONTROL POINT TABLE LOCATED IN VOLUME 0. SEE SHEET 002C105.

C. MAINTAIN MIN 1' HORIZONTAL SEPARATION BETWEEN PIPES IN SHARED TRENCH.

D. DISTRIBUTION PIPING DOWNSTREAM OF THE SUPAI TUNNEL PRV TO BE SCH 40 GALV STEEL.

KEYNOTES (X)

1. INSTALL 2.5" THREADED #24 STAINLESS STEEL MESH SCREEN AT DRAIN TERMINATION.

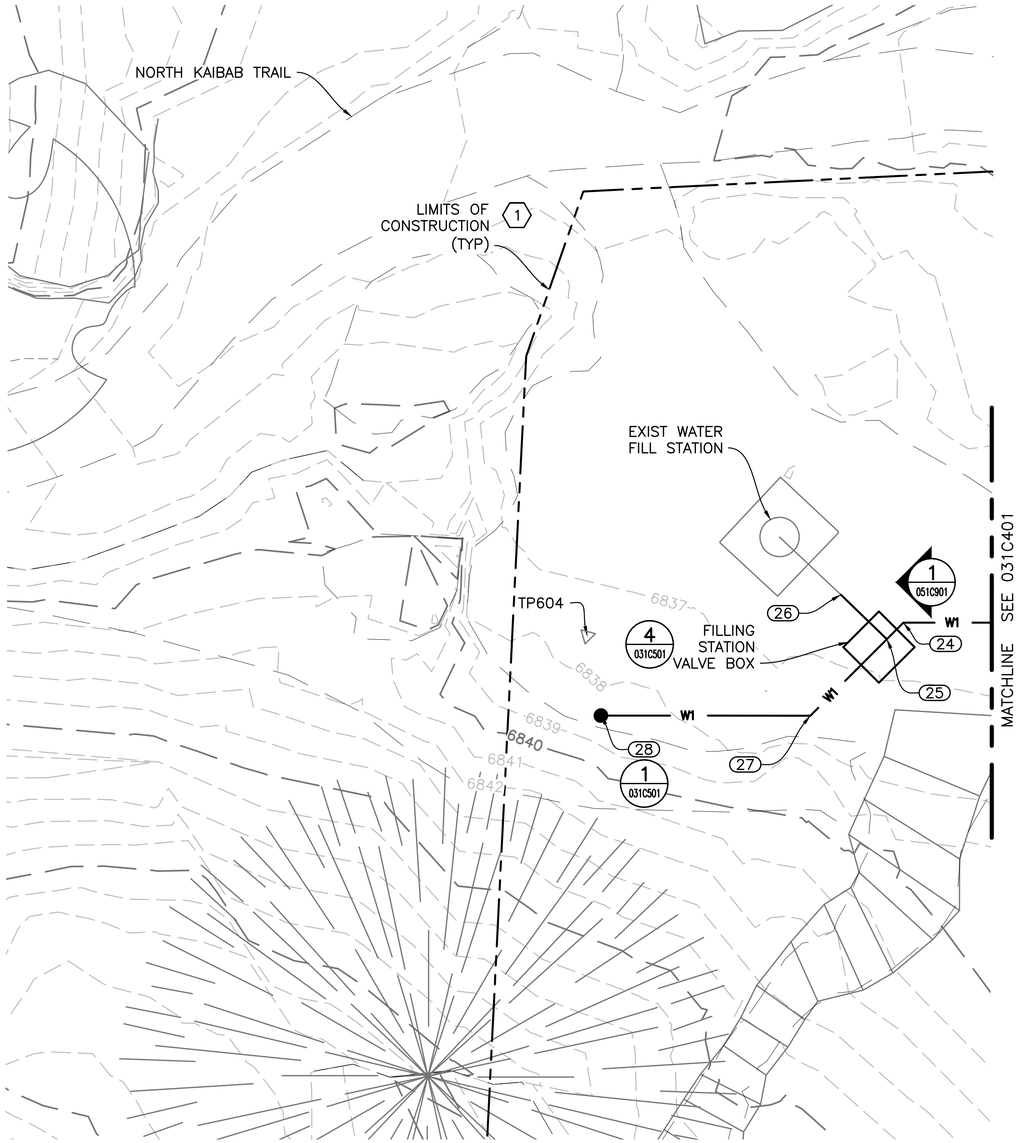
2. SEE SHEET 031C108 FOR CONSTRUCTION LIMITS TABLE.

5 0 5 10 HORIZ.
5 0 5 5 VERT.
SCALE OF FEET
H: 1"=5' V: 1"=5'

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. HUMPHREY DRAWN: C. WITTENMEIER TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 031 C401	TITLE OF SHEET FW PIPELINE SUPAI REST AREA PRV ENLARGED PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 13 OF 19	

4/16/2026 Willyard, Lisa C:\Users\LWILLYARD\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\031-C401.dwg



PLAN

LOCATION TABLE – SUPAI WATER FILLING STATIONS			
POINT NO.	NORTHING	EASTING	DESCRIPTION
24	13149302.03	1330802.05	1" 45° BEND
25	13149302.03	1330801.13	1" x 1" TEE
26	13149304.56	1330801.09	CONNECT TO EXIST 1" SERVICE LINE TO WATER FILL STATION
27	13149302.03	1330796.85	1" 45° BEND
28	13149307.90	1330790.99	INSTALL WATER FILLING STATION

GENERAL NOTES

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. SURVEY CONTROL AND CONTROL POINT TABLE LOCATED IN VOLUME 0. SEE SHEET 002C105

C. DISTRIBUTION PIPING DOWNSTREAM OF THE SUPAI TUNNEL PRV TO BE SCH 40 GALV STEEL.

KEYNOTES ⟨X⟩

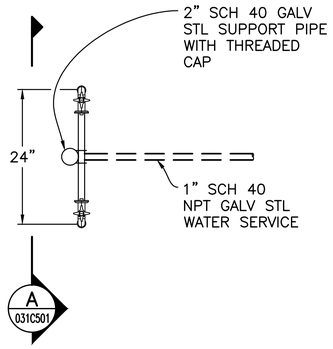
1. SEE SHEET 031C108 FOR CONSTRUCTION LIMITS TABLE.

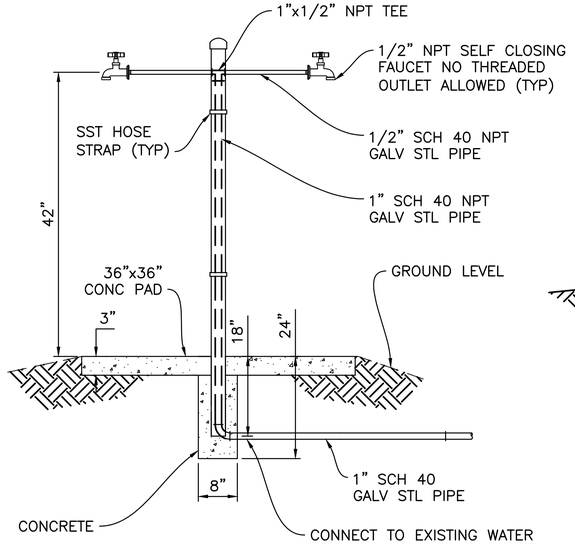
3 0 3 6
SCALE OF FEET
1"=3'

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 9-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. HUMPHREY CADD C. WITTENMEIER TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 031 C402	TITLE OF SHEET FW PIPELINE SUPAI REST AREA WATER FILLING STATIONS ENLARGED PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 14 OF 19

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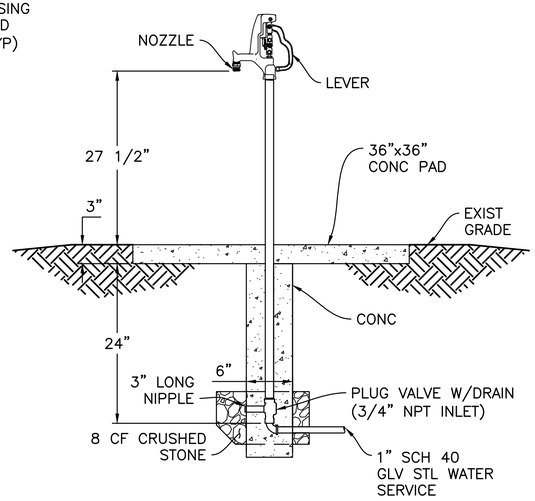
PLAN VIEW

SECTION A

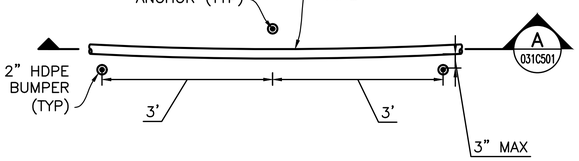
NOTES:

1. BURIED GALV STL PIPE AND FITTINGS TO BE INSTALLED WITH POLYETHYLENE ENCASEMENT WRAP.

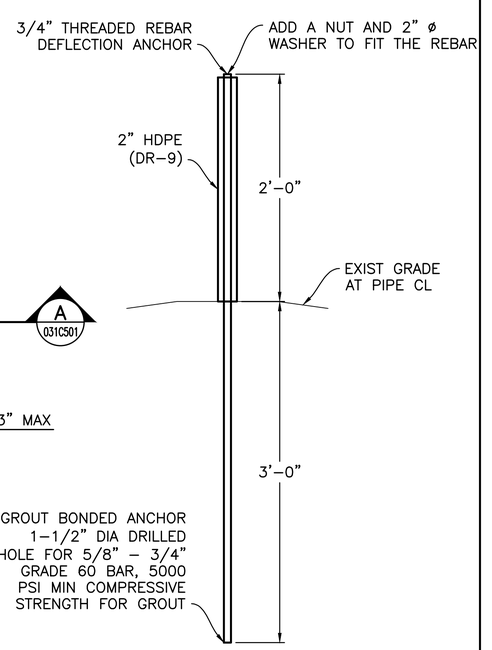
1 031C501 **WATER FILLING STATION**
NOT TO SCALE



2 031C501 **WATER SPIGOT**
NOT TO SCALE



PLAN

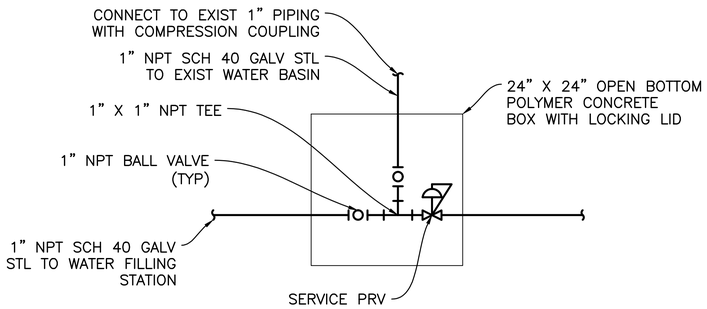


A 031C501 **ANCHOR SECTION**
NOT TO SCALE

GENERAL NOTE:

1. ALL HARDWARE TO BE GALVANIZED STEEL (G60)

3 031C501 **DEFLECTION ANCHOR**
NOT TO SCALE

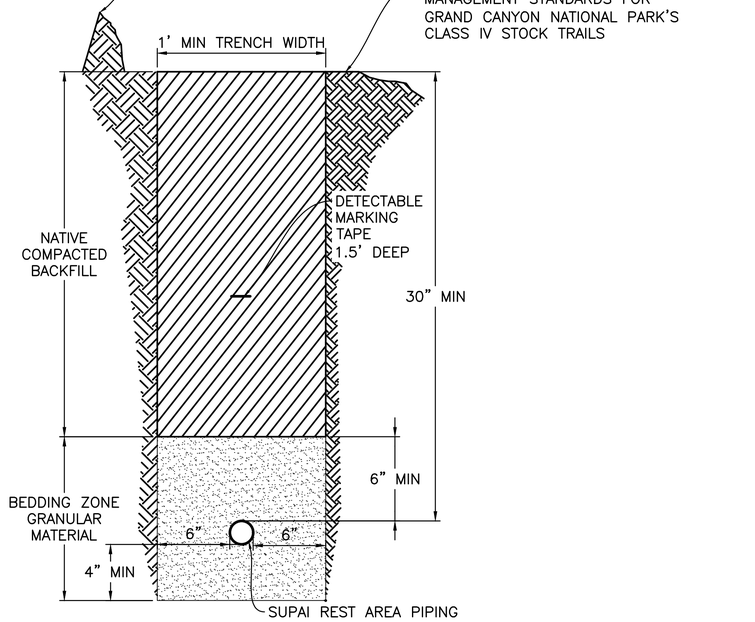


NOTES:

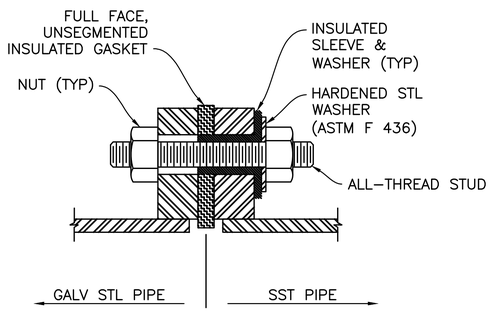
1. DEPTH OF BOX TO MATCH EXISTING PIPING + 6". CONTRACTOR TO VERIFY DEPTH PRIOR TO PROCUREMENT OF VALVE BOX.
2. CONTRACTOR TO FIELD CUT DOG HOUSE STYLE OPENINGS AS RECOMMENDED BY MFR.

INLET PRESSURE/OUTLET PRESSURE SETTING SCHEDULE	
LOCATION	SERVICE PRV (PSI)
SUPAI TUNNEL REST AREA	100/40

4 031C501 **WATER FILLING STATION VALVE BOX**
NOT TO SCALE



5 031C501 **SUPAI REST AREA TRENCH**
NOT TO SCALE



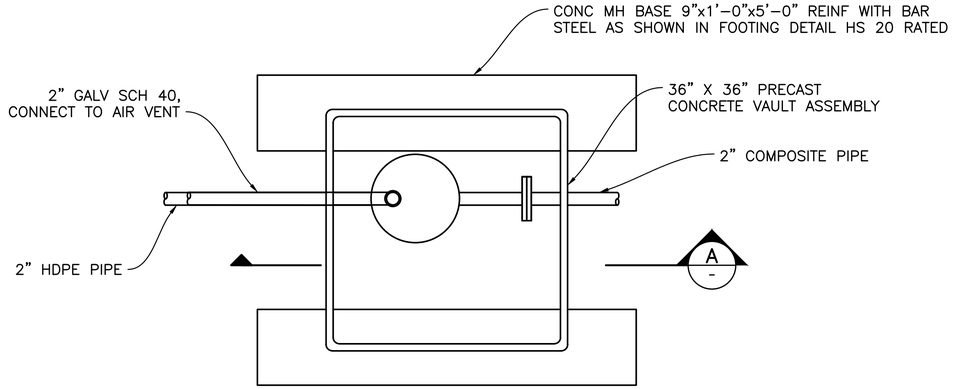
6 031C501 **INSULATING FLANGE**
NOT TO SCALE

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

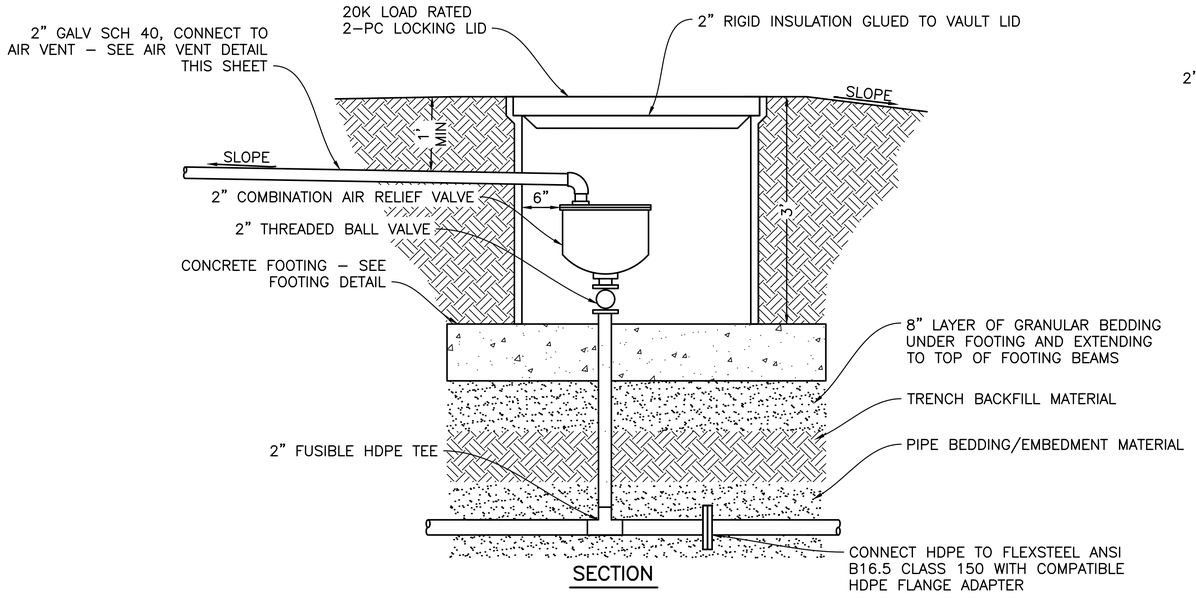
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. HUMPHREY DRAWN: C. WITTENMEIER TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 031 C501	TITLE OF SHEET FW PIPELINE WATER SPIGOT AND BOTTLE FILLING STATION DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 15 OF 19

4/16/2026 Dall, Abby C:\Users\adall\DC\ACCDocs\HDR\103505545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\031-C501.dwg



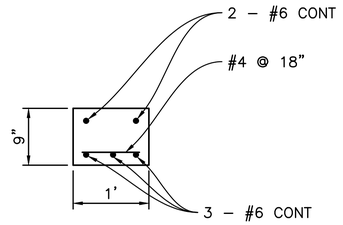
PLAN



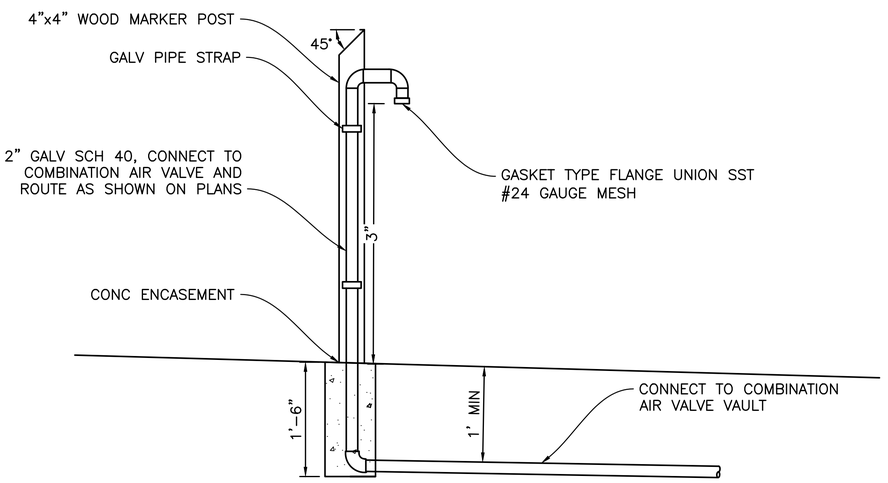
SECTION

1 / 031C502 COMBINATION AIR RELEASE/VACUUM RELIEF VALVE ASSEMBLY DETAIL

NOT TO SCALE



FOOTING DETAIL

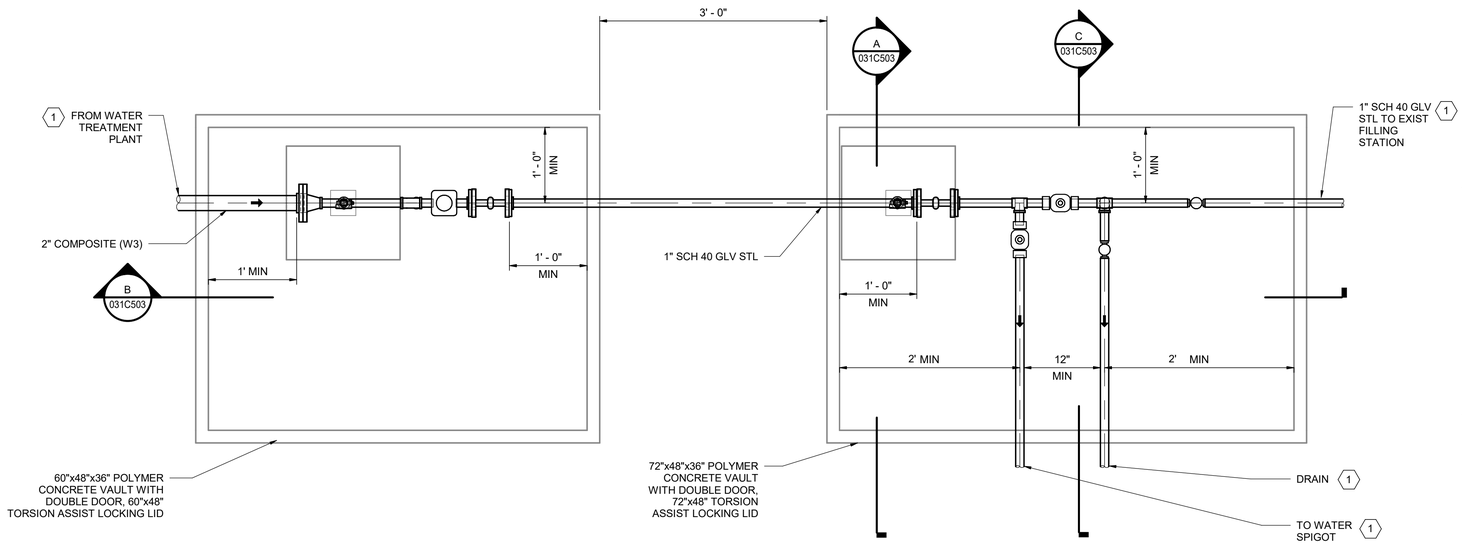


AIR VENT DETAIL

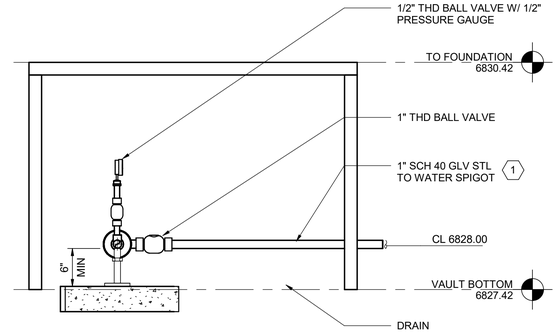
Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. HUMPHREY CADD C. CONFER TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 031 C502	TITLE OF SHEET FW PIPELINES AIR RELIEF VALVE VAULT ASSEMBLY PLAN AND SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 16 OF 19

4/16/2026 Willyard, Lisa C:\Users\LWILLYARD\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\031-C502.dwg



1 031C503 PRV VAULT PLAN
SCALE A



A 031C503 SECTION
SCALE A

GENERAL NOTES

A. INLET PRESSURES SHOWN IN THE SCHEDULE ARE BASED ON NORMAL OPERATING FLOW. OUTLET PRESSURE SETTINGS ARE THE SAME FOR ALL FLOW CONDITIONS.

B. CONTRACTOR / MFR TO EVALUATE & DETERMINED PRV SIZING AND CONNECTIONS FOR THE PRODUCT BEING SUPPLIED BASED ON REQUIREMENTS SHOWN IN SCHEDULE.

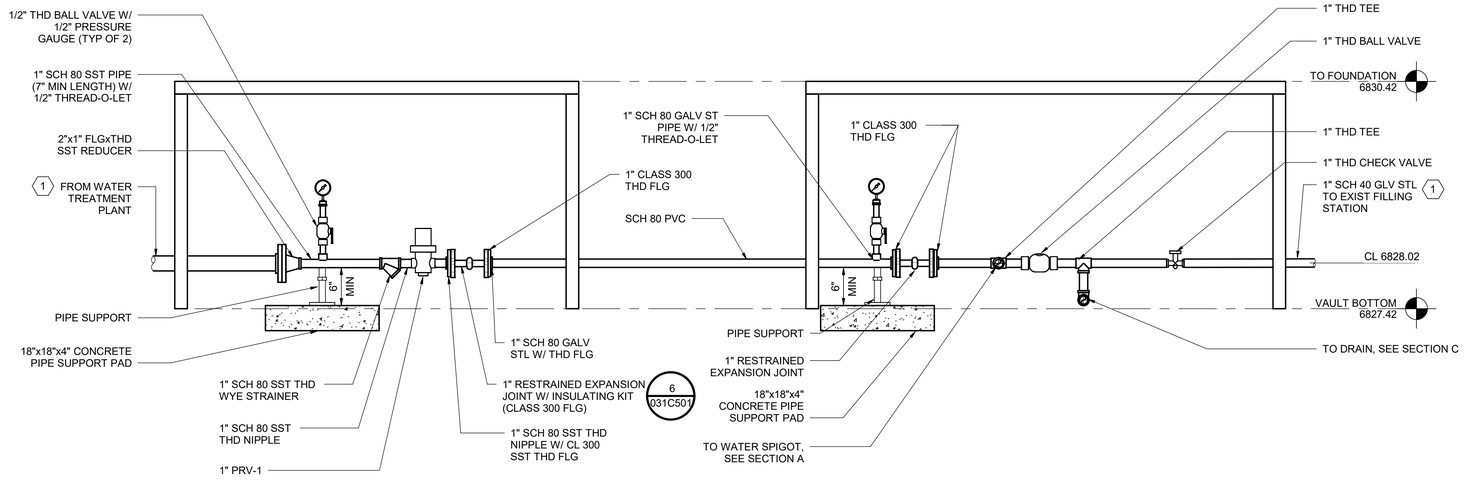
C. PIPE, VALVE & FITTING SIZE TO MATCH SIZE OF PRV SUPPLIED.

KEY NOTES X

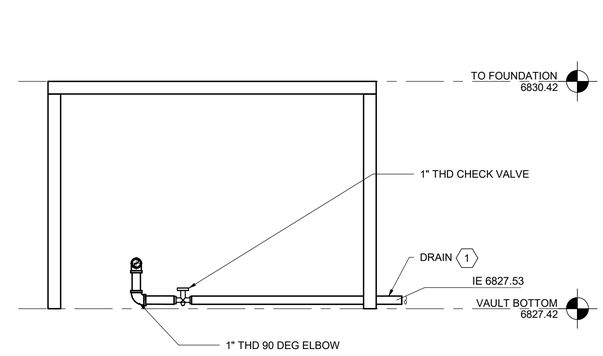
1. SEE 031C401 FOR CONTINUATION OF PIPING.

INLET PRESSURE/OUTLET PRESSURE SETTING SCHEDULE

LOCATION	PRV 1 (PSI)
SUPAI TUNNEL REST AREA	690/100



B 031C503 SECTION
SCALE A



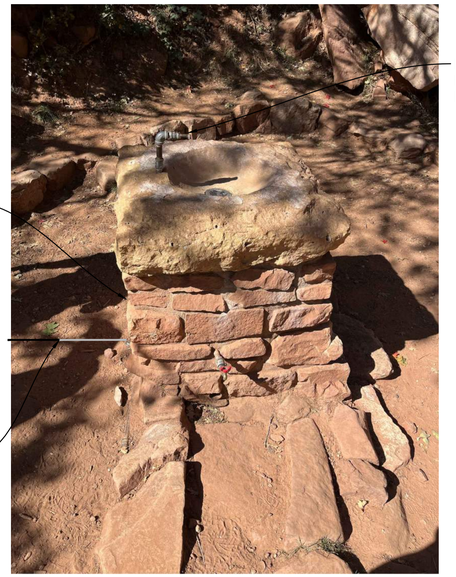
C 031C503 SECTION
SCALE A

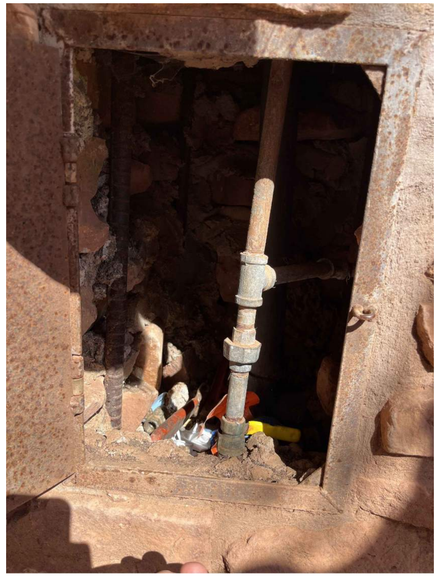
SCALE A 1 0 1 2 1" = 1'-0"
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY 5-1-2026 Registered ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. HUMPHREY DRAWN: C. CONFER TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 031 C503	TITLE OF SHEET FW PIPELINE RESTHOUSE VAULT ASSEMBLY PLAN AND SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 17 OF 19

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ENLARGED PHOTO

1 / 031C901 EXISTING WATER FILLING STATION MODIFICATION

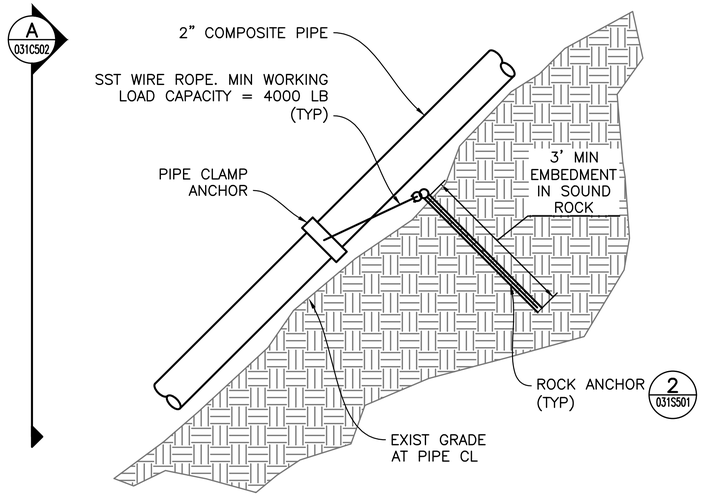
NOT TO SCALE

4/29/2026 Race, Sara C:\Users\sracE\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\031-C901.dwg

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Not Signed ARIZONA, U.S.A.

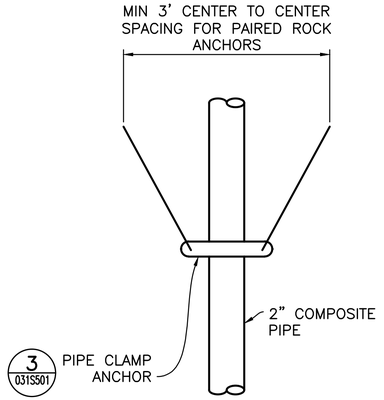
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. HUMPHREY CADD C. WITTENMEIER TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 031 C901	TITLE OF SHEET FW PIPELINE PHOTO DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 18 OF 19

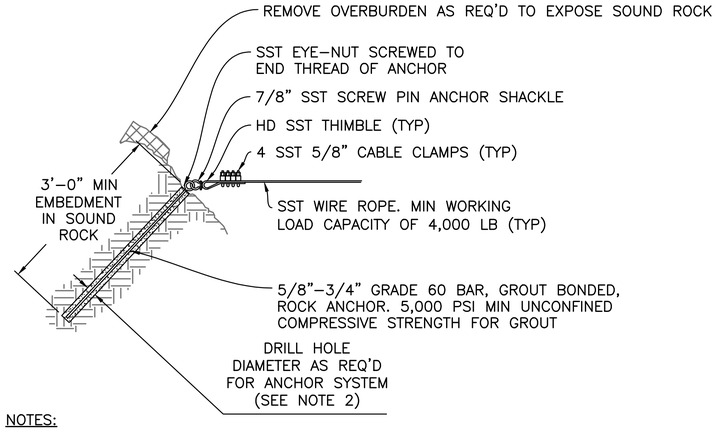


NOTES:

1. MAX DISTANCE BETWEEN PIPE ANCHOR POINTS SHALL BE 275 FT OF PIPE LENGTH.
2. LOCATE ROCK ANCHOR POINTS AT EACH LOCATION PRIOR TO INSTALLING ANCHORS AND STEEL CABLES. STEEL CABLES SHALL BE PRECUT WITH THIMBLES AND SWAGED ENDS AND INSTALLED WITH HARDWARE SHOWN. ALL WIRE SHALL BE TAUT USING TURNBUCKLES AS SHOWN.
3. SLOPE OF WIRE ROPE SHALL BE NO LESS THAN 2V:1H.
4. ALL STEEL HARDWARE SHALL BE STAINLESS TYPE 304 OR 316 UNLESS OTHERWISE NOTED.



A 031C502 ROCK ANCHOR SECTION
NOT TO SCALE

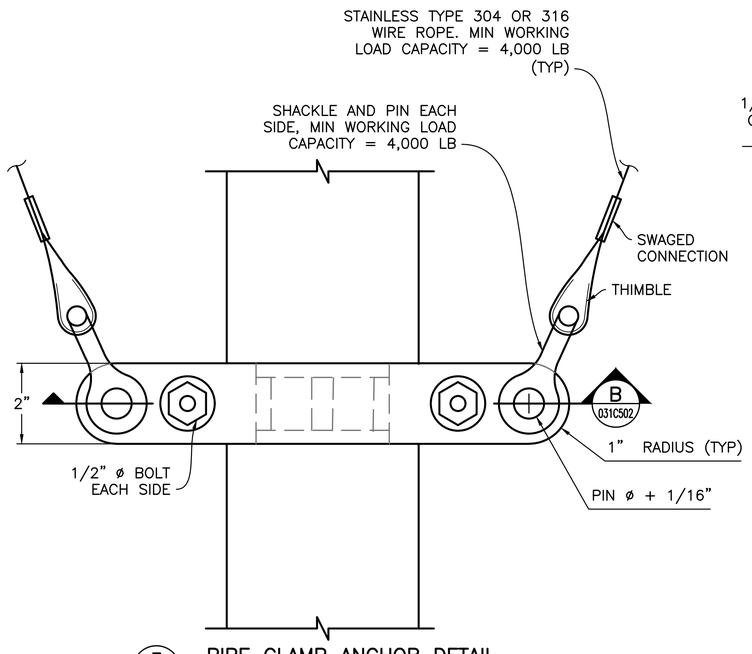


NOTES:

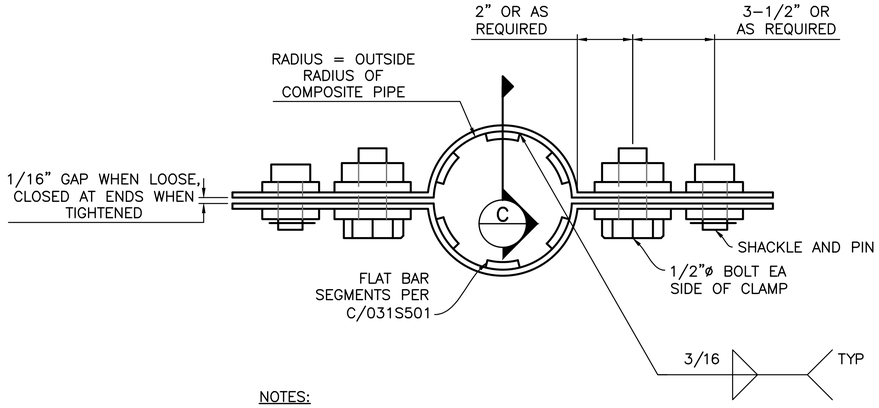
1. DRILL HOLE USING ROTARY PERCUSSION EQUIPMENT SUCH THAT BORE DRILLED HOLE SIDE WALLS ARE ROUGH AND NOT SMOOTH, AS RECOMMENDED BY RESIN GROUT MANUFACTURER.
2. DRILLED HOLE DIAMETER SHALL BE 1-1/2" ø.
3. GROUT BONDED ROCK ANCHORS TO BE DESIGNED AND SPECIFIED BY CONTRACTOR'S ENGINEER. ANCHORS SHALL HAVE A MIN COMBINED CAPACITY OF 8000 LB APPLIED IN ANY DIRECTION ON ANCHOR GROUP AND INDIVIDUAL ANCHOR CAPACITY OF 4,000 LB SHEAR AND 4,000 LB TENSION.
4. ALL STEEL HARDWARE SHALL BE STAINLESS TYPE 304 OR 316 UNLESS OTHERWISE NOTED.

2 031S501 ROCK ANCHOR
NOT TO SCALE

1 031S501 PIPE ANCHOR DETAIL
NOT TO SCALE



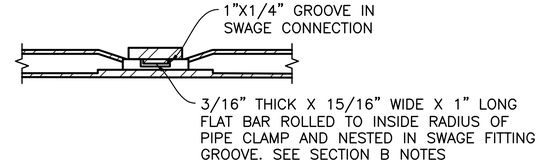
3 031S501 PIPE CLAMP ANCHOR DETAIL
NOT TO SCALE



NOTES:

1. PROVIDE MOCK-UP OF PIPE CLAMP FOR APPROVAL BY CONTRACTING OFFICER PRIOR TO FABRICATION OF PIPE CLAMPS.
2. MOCK-UP SHALL BE LOAD TESTED BY CONTRACTOR FOR CAPACITY OF 8,000 LB FOR 24 HOURS. CONTRACTOR TO SUBMIT LOAD TEST PROCEDURE FOR APPROVAL PRIOR TO TESTING. CONTRACTING OFFICER SHALL BE INVITED TO WITNESS LOAD TEST. FINAL TEST REPORT SHALL BE SUBMITTED TO CONTRACTING OFFICER WITHIN 7 DAYS.

B 031S501 PIPE CLAMP SECTION
NOT TO SCALE



C 031S501 PIPE CLAMP SECTION
NOT TO SCALE

Registered Professional Engineer CERTIFICATE NO. 22198 NICHOLAS J. LAFRONZ Date Signed 6-1-26 ARIZONA, U.S.A.

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. HUMPHREY CADD C. WITTENMEIER TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 031 S501	TITLE OF SHEET FW PIPELINE PIPELINE ANCHOR DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 19 OF 19

4/17/2026 Dell, Abby C:\Users\adall\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\STRUC\031-S501.dwg